3 Scattering Theory

The general principles of relativistic quantum mechanics described in the previous chapter have so far been applied here only to states of a single stable particle. Such one-particle states by themselves are not very exciting — it is only when two or more particles interact with each other that anything interesting can happen. But experiments do not generally follow the detailed course of events in particle interactions. Rather, the paradigmatic experiment (at least in nuclear or elementary particle physics) is one in which several particles approach each other from a macroscopically large distance, and interact in a microscopically small region, after which the products of the interaction travel out again to a macroscopically large distance. The physical states before and after the collision consist of particles that are so far apart that they are effectively non-interacting, so they can be described as direct products of the one-particle states discussed in the previous chapter. In such an experiment, all that is measured is the probability distribution, or 'cross-sections', for transitions between the initial and final states of distant and effectively non-interacting particles. This chapter will outline the formalism [1] used for calculating these probabilities and cross-sections.

3.1 'In' and 'Out' States

A state consisting of several non-interacting particles may be regarded as one that transforms under the Poincaré (inhomogeneous Lorentz) group as a direct product of one-particle states. To label the one-particle states we use their four-momenta p^μ, spin z-component (or, for massless particles, helicity) σ, and, since we now may be dealing with more than one species of particle, an additional discrete label n for the particle type, which includes a specification of its mass, spin, charge, etc. The general transformation rule is

$$\begin{aligned} U(\Lambda, a)|p_1, \sigma_1, n_1;\, p_2, \sigma_2, n_2;\, \ldots\rangle =\; & \exp\big[-ia_\mu\big(p_1^\mu + p_2^\mu + \cdots\big)\big] \sqrt{\frac{(\Lambda p_1)^0(\Lambda p_2)^0\cdots}{p_1^0 p_2^0\cdots}} \\ & \times \sum_{\sigma_1'\sigma_2'\cdots} D^{(j_1)}_{\sigma_1'\sigma_1}\big(W(\Lambda, p_1)\big) D^{(j_2)}_{\sigma_2'\sigma_2}\big(W(\Lambda, p_2)\big)\cdots \\ & \times |\Lambda p_1, \sigma_1', n_1;\, \Lambda p_2, \sigma_2', n_2;\, \ldots\rangle. \end{aligned} \tag{3.1.1}$$

where $W(\Lambda, p)$ is the Wigner rotation (2.5.10), and $D^{(j)}_{\sigma'\sigma}(W)$ are the conventional $(2j+1)$-dimensional unitary matrices representing the three-dimensional rotation group. (This is for massive particles; for any massless particle, the matrix $D^{(j)}_{\sigma'\sigma}(W(\Lambda, p))$ is replaced with $\delta_{\sigma'\sigma}\exp(i\sigma\theta(\Lambda, p))$, where θ is the angle defined by Eq. (2.5.43).) The states are normalized as in Eq. (2.5.19):

$$\begin{aligned} & \big\langle p_1', \sigma_1', n_1';\, p_2', \sigma_2', n_2';\, \ldots \,\big|\, p_1, \sigma_1, n_1;\, p_2, \sigma_2, n_2;\, \ldots\big\rangle \\ & \quad = \delta^3(\mathbf{p}_1' - \mathbf{p}_1)\delta_{\sigma_1'\sigma_1}\delta_{n_1'n_1}\delta^3(\mathbf{p}_2' - \mathbf{p}_2)\delta_{\sigma_2'\sigma_2}\delta_{n_2'n_2}\cdots \\ & \qquad \pm\text{permutations}. \end{aligned} \tag{3.1.2}$$

with the term '$\pm$ permutations' included to take account of the possibility that it is some permutation of the particle types $n_1', n_2', \ldots$ that are of the same species of the particle types $n_1, n_2, \ldots$. (As discussed more fully in Chapter 4, its sign is -1 if this permutation includes an odd permutation of half-integer spin particles, and otherwise $+1$. This will not be important in the work of the present chapter.)

We often use an abbreviated notation, letting one Greek letter, say α, stand for the whole collection p_1, σ_1, n_1; p_2, σ_2, n_2; $\ldots$. In this notation, Eq. (3.1.2) is written simply

$$\langle \alpha' \mid \alpha \rangle = \delta(\alpha' - \alpha), \tag{3.1.3}$$

with $\delta(\alpha' - \alpha)$ standing for the sum of products of delta functions and Kronecker deltas appearing on the right-hand side of Eq. (3.1.2). Also, in summing over states, we write

$$\int d\alpha \cdots \equiv \sum_{n_1 \sigma_1 n_2 \sigma_2 \cdots} \int d^3p_1 \, d^3p_2 \cdots . \tag{3.1.4}$$

In particular, the completeness relation for states normalized as in Eq. (3.1.3) reads

$$|\Psi\rangle = \int d\alpha \, |\alpha\rangle \langle \alpha \mid \Psi \rangle. \tag{3.1.5}$$

The transformation rule (3.1.1) is only possible for particles that for one reason or another are not interacting. Setting $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu$ and $a^\mu = (\tau, 0, 0, 0)$, for which $U(\Lambda, a) = \exp(iH\tau)$, Eq. (3.1.1) requires among other things that $|\alpha\rangle$ be an energy eigenstate,

$$H|\alpha\rangle = E_\alpha |\alpha\rangle, \tag{3.1.6}$$

with an energy equal to the sum of the one-particle energies,

$$E_\alpha = p_1^0 + p_2^0 + \cdots , \tag{3.1.7}$$

and with no interaction terms, terms that would involve more than one particle at a time.

On the other hand, the transformation rule (3.1.1) does apply in scattering processes at times $t \to \mp\infty$. As explained at the beginning of this chapter, in the typical scattering experiment we start with particles at time $t \to -\infty$ so far apart that they are not yet interacting, and end with particles at $t \to +\infty$ so far apart that they have ceased interacting. We therefore have not one but two sets of states that transform as in Eq. (3.1.1): the 'in' and 'out' states[1] $|\alpha\rangle_{\text{in}}$ and $|\alpha\rangle_{\text{out}}$ will be found to contain the particles described by the label α if observations are made at $t \to -\infty$ or $t \to +\infty$, respectively.

Note how this definition is framed. To maintain manifest Lorentz invariance, in the formalism we are using here, state-vectors do not change with time — a state-vector $|\Psi\rangle$ describes the whole spacetime history of a system of particles. (This is known as the *Heisenberg picture*, in distinction with the Schrödinger picture, where the operators are constant and the states change with time.) Thus we do *not* say that $|\alpha\rangle_{\text{in}}$ and $|\alpha\rangle_{\text{out}}$ are the limits at $t \to -\infty$ and $t \to +\infty$ of a time-dependent state-vector $|\Psi(t)\rangle$.

However, implicit in the definition of the states is a choice of the inertial frame from which the observer views the system; different observers see *equivalent* state-vectors, but

[1] The labels '+' and '−' for 'in' and 'out' states may seem backward, but they seem to have become traditional. They arise from the signs in Eq. (3.1.16).

not the *same* state-vector. In particular, suppose that a standard observer O sets his or her clock so that $t = 0$ is at some time during the collision process, while some other observer O' at rest with respect to the first uses a clock set so that $t' = 0$ is at a time $t = \tau$; that is, the two observers' time coordinates are related by $t' = t - \tau$. Then if O sees the system to be in a state $|\Psi\rangle$, O' will see the system in a state $U(\mathbf{1}, -\tau)|\Psi\rangle = \exp(-iH\tau)|\Psi\rangle$. Thus the appearance of the state long before or long after the collision (in whatever basis is used by O) is found by applying a time-translation operator $\exp(-iH\tau)$ with $\tau \to -\infty$ or $\tau \to +\infty$, respectively. Of course, if the state is really an energy eigenstate, then it cannot be localized in time — the operator $\exp(-iH\tau)$ yields an inconsequential phase factor $\exp(-iE_\alpha\tau)$. Therefore, we must consider wave-packets, superpositions $\int d\alpha\, g(\alpha)|\alpha\rangle$ of states, with an amplitude $g(\alpha)$ that is non-zero and smoothly varying over some finite range ΔE of energies. The 'in' and 'out' states are defined so that the superposition

$$\exp(-iH\tau) \int d\alpha\, g(\alpha)|\alpha\rangle_{\text{in/out}} = \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle_{\text{in/out}}$$

has the appearance of a corresponding superposition of free-particle states for $\tau \ll -1/\Delta E$ or $\tau \gg +1/\Delta E$, respectively.

To make this concrete, suppose we can divide the time-translation generator H into two terms, a free-particle Hamiltonian H_0 and an interaction H_{int},

$$H = H_0 + H_{\text{int}}, \tag{3.1.8}$$

in such a way that H_0 has eigenstates $|\alpha\rangle$ that have the same appearance as the eigenstates $|\alpha\rangle_{\text{in}}$ and $|\alpha\rangle_{\text{out}}$ of the complete Hamiltonian,

$$H_0|\alpha\rangle = E_\alpha|\alpha\rangle, \tag{3.1.9}$$

$$\langle\alpha' \mid \alpha\rangle = \delta(\alpha' - \alpha). \tag{3.1.10}$$

Note that H_0 is assumed here to have the same spectrum as the full Hamiltonian H. This requires that the masses appearing in H_0 be the physical masses that are actually measured, which are not necessarily the same as the 'bare' mass terms appearing in H; the difference if there is any must be included in the interaction H_{int}, not H_0. Also, any relevant bound states in the spectrum of H should be introduced into H_0 as if they were elementary particles.[2]

The 'in' and 'out' states can now be defined as eigenstates of H, not H_0,

$$H|\alpha\rangle_{\text{in/out}} = E_\alpha|\alpha\rangle_{\text{in/out}}, \tag{3.1.11}$$

which satisfy the condition

$$\begin{aligned} \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle_{\text{in}} &\longrightarrow \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle \qquad (\tau \to -\infty), \\ \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle_{\text{out}} &\longrightarrow \int d\alpha\, e^{-iE_\alpha\tau} g(\alpha)|\alpha\rangle \qquad (\tau \to +\infty). \end{aligned} \tag{3.1.12}$$

[2] Alternatively, in non-relativistic problems we can include the binding potential in H_0. In the application of this method to 'rearrangement collisions,' where some bound states appear in the initial state but not the final state, or vice-versa, one must use a different split of H into H_0 and H_{int} in the initial and final states.

Eq. (3.1.12) can be rewritten as the requirement that:

$$\exp(-iH\tau) \int d\alpha\, g(\alpha) |\alpha\rangle_{\text{in/out}} \longrightarrow \exp(-iH_0\tau) \int d\alpha\, g(\alpha) |\alpha\rangle$$

for $\tau \to -\infty$ or $\tau \to +\infty$, respectively. This is sometimes rewritten as a formula for the 'in' and 'out' states:

$$\begin{aligned} |\alpha\rangle_{\text{in}} &= \Omega(-\infty)|\alpha\rangle, \\ |\alpha\rangle_{\text{out}} &= \Omega(+\infty)|\alpha\rangle, \end{aligned} \tag{3.1.13}$$

where

$$\Omega(\tau) \equiv \exp(+iH\tau)\exp(-iH_0\tau). \tag{3.1.14}$$

However, it should be kept in mind that $\Omega(\mp\infty)$ in Eq. (3.1.13) gives meaningful results only when acting on a smooth superposition of energy eigenstates.

One immediate consequence of the definition (3.1.12) is that the 'in' and 'out' states are normalized just like the free-particle states. To see this, note that since the left-hand side of Eq. (3.1.12) is obtained by letting the unitary operator $\exp(-iH\tau)$ act on a time-independent state, its norm is independent of time, and therefore equals the norm of its limit for $\tau \to \mp\infty$, i.e., the norm of the right-hand side of Eq. (3.1.12):

$$\begin{aligned} &\int d\alpha\, d\beta\, \exp\bigl(-i(E_\alpha - E_\beta)\tau\bigr) g(\alpha) g^*(\beta)\, {}_{\text{in/out}}\bigl\langle \beta \mid \alpha \bigr\rangle_{\text{in/out}} \\ &\qquad = \int d\alpha\, d\beta\, \exp\bigl(-i(E_\alpha - E_\beta)\tau\bigr) g(\alpha) g^*(\beta) \langle \beta \mid \alpha \rangle. \end{aligned}$$

Since this is supposed to be true for all smooth functions $g(\alpha)$, the scalar products must be equal:

$${}_{\text{in/out}}\bigl\langle \beta \mid \alpha \bigr\rangle_{\text{in/out}} = \langle \beta \mid \alpha \rangle = \delta(\beta - \alpha). \tag{3.1.15}$$

It is useful for some purposes to have an explicit though formal solution of the energy eigenvalue equation (3.1.11) satisfying the conditions (3.1.12). For this purpose, write Eq. (3.1.11) as

$$(E_\alpha - H_0)|\alpha\rangle_{\text{in/out}} = H_{\text{int}}|\alpha\rangle_{\text{in/out}}.$$

The operator $E_\alpha - H_0$ is not invertible; it annihilates not only the free-particle state $|\alpha\rangle$, but also the continuum of other free-particle states $|\beta\rangle$ of the same energy. Since the 'in' and 'out' states become just $|\alpha\rangle$ for $H_{\text{int}} \to 0$, we tentatively write the formal solutions as $|\alpha\rangle$ plus a term proportional to H_{int}:

$$\begin{aligned} |\alpha\rangle_{\text{in}} &= |\alpha\rangle + (E_\alpha - H_0 + i\epsilon)^{-1} H_{\text{int}} |\alpha\rangle_{\text{in}}, \\ |\alpha\rangle_{\text{out}} &= |\alpha\rangle + (E_\alpha - H_0 - i\epsilon)^{-1} H_{\text{int}} |\alpha\rangle_{\text{out}}, \end{aligned} \tag{3.1.16}$$

or, expanding in a complete set of free-particle states,

$$\begin{aligned} |\alpha\rangle_{\text{in}} &= |\alpha\rangle + \int d\beta\, \frac{T^{\text{in}}_{\beta\alpha}|\beta\rangle}{E_\alpha - E_\beta + i\epsilon}, \\ |\alpha\rangle_{\text{out}} &= |\alpha\rangle + \int d\beta\, \frac{T^{\text{out}}_{\beta\alpha}|\beta\rangle}{E_\alpha - E_\beta - i\epsilon}, \end{aligned} \tag{3.1.17}$$

$$T^{\rm in/out}_{\beta\alpha} \equiv \langle\beta|H_{\rm int}|\alpha\rangle_{\rm in/out}, \tag{3.1.18}$$

with ϵ a positive infinitesimal quantity, inserted to give meaning to the reciprocal of $E_\alpha - H_0$. These are known as the *Lippmann–Schwinger equations* [1a]. We shall use Eq. (3.1.17) at the end of the next section to give a slightly less unrigorous proof of the orthonormality of the 'in' and 'out' states.

It remains to be shown that Eq. (3.1.17), with a $+i\epsilon$ or $-i\epsilon$ in the denominator, satisfies the condition (3.1.12) for an 'in' or an 'out' state, respectively. For this purpose, consider the superpositions

$$|\Psi_g(t)\rangle_{\rm in/out} \equiv \int d\alpha\, e^{-iE_\alpha t} g(\alpha)|\alpha\rangle_{\rm in/out}, \tag{3.1.19}$$

$$|\Phi_g(t)\rangle \equiv \int d\alpha\, e^{-iE_\alpha t} g(\alpha)|\alpha\rangle. \tag{3.1.20}$$

We want to show that $|\Psi_g(t)\rangle_{\rm in}$ and $|\Psi_g(t)\rangle_{\rm out}$ approach $|\Phi_g(t)\rangle$ for $t \to -\infty$ and $t \to +\infty$, respectively. Using Eq. (3.1.17) in Eq. (3.1.19) gives

$$|\Psi_g(t)\rangle_{\rm in/out} = |\Phi_g(t)\rangle + \int d\alpha \int d\beta\, \frac{e^{-iE_\alpha t} g(\alpha) T^{\rm in/out}_{\beta\alpha}|\beta\rangle}{E_\alpha - E_\beta \pm i\epsilon}, \tag{3.1.21}$$

where the upper and lower signs refer to 'in' and 'out', respectively. Let us recklessly interchange the order of integration, and consider first the integrals

$$\mathcal{I}^{\rm in/out}_\beta \equiv \int d\alpha\, \frac{e^{-iE_\alpha t} g(\alpha) T^{\rm in/out}_{\beta\alpha}}{E_\alpha - E_\beta \pm i\epsilon}.$$

For $t \to -\infty$, we can close the contour of integration for the energy variable E_α in the upper half-plane with a large semi-circle, with the contribution from this semi-circle killed by the factor $\exp(-iE_\alpha t)$, which is exponentially small for $t \to -\infty$ and $\operatorname{Im} E_\alpha > 0$. The integral is then given by a sum over the singularities of the integrand in the upper half-plane. The functions $g(\alpha)$ and $T^{\rm in/out}_{\beta\alpha}$ may, in general, be expected to have some singularities at values of E_α with finite positive imaginary parts, but just as for the large semi-circle, their contribution is exponentially damped for $t \to -\infty$. (Specifically, $-t$ must be much greater than both the time-uncertainty in the wave-packet $g(\alpha)$ and the duration of the collision, which respectively govern the location of the singularities of $g(\alpha)$ and $T^{\rm in/out}_{\beta\alpha}$ in the complex E_α plane.) This leaves the singularity in $(E_\alpha - E_\beta \pm i\epsilon)^{-1}$, which is in the upper half-plane for $\mathcal{I}^{\rm out}_\beta$ but not $\mathcal{I}^{\rm in}_\beta$. We conclude then that $\mathcal{I}^{\rm in}_\beta$ vanishes for $t \to -\infty$. In the same way, for $t \to +\infty$ we must close the contour of integration in the lower half-plane, and so $\mathcal{I}^{\rm out}_\beta$ vanishes in this limit. We conclude that $|\Psi_g(t)\rangle_{\rm in/out}$ approaches $|\Phi_g(t)\rangle$ for $t \to \mp\infty$, in agreement with the defining condition (3.1.12).

* * *

For future use, we note a convenient representation of the factor $(E_\alpha - E_\beta \pm i\epsilon)^{-1}$ in Eq. (3.1.17). In general, we can write

$$(E \pm i\epsilon)^{-1} = \frac{\Pi_\epsilon}{E} \mp i\pi\delta_\epsilon(E), \tag{3.1.22}$$

where

$$\frac{\Pi_\epsilon}{E} \equiv \frac{E}{E^2+\epsilon^2}, \tag{3.1.23}$$

$$\delta_\epsilon(E) \equiv \frac{\epsilon}{\pi(E^2+\epsilon^2)}. \tag{3.1.24}$$

The function (3.1.23) is just $1/E$ for $|E| \gg \epsilon$, and vanishes for $E \to 0$, so for $\epsilon \to 0$ it behaves just like the 'principal value function' Π/E, which allows us to give meaning to integrals of $1/E$ times any smooth function of E, by excluding an infinitesimal interval around $E = 0$. The function (3.1.24) is of order ϵ for $|E| \gg \epsilon$, and gives unity when integrated over all E, so in the limit $\epsilon \to 0$ it behaves just like the familiar delta function $\delta(E)$. With this understanding, we can drop the label ϵ in Eq. (3.1.22), and write simply

$$(E \pm i\epsilon)^{-1} = \frac{\Pi}{E} \mp i\pi\delta(E). \tag{3.1.25}$$

3.2 The S-Matrix

An experimentalist generally prepares a state to have a definite particle content at $t \to -\infty$, and then measures what this state looks like at $t \to +\infty$. If the state is prepared to have a particle content α for $t \to -\infty$, then it is the 'in' state $|\alpha\rangle_{\rm in}$, and if it is found to have the particle content β at $t \to +\infty$, then it is the 'out' state $|\beta\rangle_{\rm out}$. The probability amplitude for the transition $\alpha \to \beta$ is thus the scalar product

$$S_{\beta\alpha} = {}_{\rm out}\langle \beta \mid \alpha \rangle_{\rm in}. \tag{3.2.1}$$

This array of complex amplitudes is known as the S-matrix [2]. If there were no interactions then 'in' and 'out' states would be the same, and then $S_{\beta\alpha}$ would be just $\delta(\alpha-\beta)$. The rate for a reaction $\alpha \to \beta$ is thus proportional to $|S_{\beta\alpha} - \delta(\alpha-\beta)|^2$. We shall see in detail in Section 3.4 what $S_{\beta\alpha}$ has to do with measured rates and cross-sections.

Perhaps it should be stressed that 'in' and 'out' states do not inhabit two different Hilbert spaces. They differ only in how they are labelled: by their appearance either at $t \to -\infty$ or $t \to +\infty$. Any 'in' state can be expanded as a sum of 'out' states, with expansion coefficients given by the S-matrix (3.2.1).

Since $S_{\beta\alpha}$ is the matrix connecting two sets of orthonormal states, it must be unitary. To see this in greater detail, apply the completeness relation (3.1.5) to the 'out' states, and write

$$\int d\beta\, S^*_{\beta\gamma} S_{\beta\alpha} = \int d\beta\, {}_{\rm in}\langle \gamma \mid \beta \rangle_{\rm out}\, {}_{\rm out}\langle \beta \mid \alpha \rangle_{\rm in} = {}_{\rm in}\langle \gamma \mid \alpha \rangle_{\rm in}.$$

Using (3.1.15), this gives

$$\int d\beta\, S^*_{\beta\gamma} S_{\beta\alpha} = \delta(\gamma-\alpha), \tag{3.2.2}$$

or in brief, $S^\dagger S = \mathbf{1}$. In the same way, completeness for the 'in' states gives[3]

$$\int d\beta\, S_{\gamma\beta} S^*_{\alpha\beta} = \delta(\gamma-\alpha), \tag{3.2.3}$$

[3]An alternative proof is given at the end of this section. Note that for infinite 'matrices,' the unitarity conditions $S^\dagger S = \mathbf{1}$ and $SS^\dagger = \mathbf{1}$ are not equivalent.

or, in other words, $SS^\dagger = \mathbf{1}$.

It is often convenient instead of dealing with the S-matrix to work with an operator S, defined to have matrix elements between *free-particle* states equal to the corresponding elements of the S-matrix:

$$\langle\beta|S|\alpha\rangle \equiv S_{\beta\alpha}. \tag{3.2.4}$$

The explicit though highly formal expression (3.1.13) for the 'in' and 'out' states yields a formula for the S-operator:

$$S = \Omega(+\infty)^\dagger\Omega(-\infty) = U(+\infty, -\infty), \tag{3.2.5}$$

where

$$U(\tau, \tau_0) \equiv \Omega(\tau)^\dagger\Omega(\tau_0) = \exp(iH_0\tau)\exp\big(-iH(\tau-\tau_0)\big)\exp(-iH_0\tau_0). \tag{3.2.6}$$

This will be used in the next section to examine the Lorentz invariance of the S-matrix, and in Sec. 3.5 to derive a formula for the S-matrix in time-dependent perturbation theory.

The methods of the previous section can be used to derive a useful alternative formula for the S-matrix. Let's return to Eq. (3.1.21) for the 'in' state $|\Psi_g\rangle_{\rm in}$, but this time take $t \to +\infty$. We must now close the contour of integration for E_α in the *lower* half-E_α-plane, and although as before the singularities in $T^{\rm in}_{\beta\alpha}$ and $g(\alpha)$ make no contribution for $t \to +\infty$, we now do pick up a contribution from the singular factor $(E_\alpha - E_\beta + i\epsilon)^{-1}$. The contour runs from $E_\alpha = -\infty$ to $E_\alpha = +\infty$, and then back to $E_\alpha = -\infty$ on a large semi-circle in the lower half-plane, so it circles the singularity in a *clockwise* direction. By the method of residues, this contribution to the integral over E_α is given by the value of the integrand at $E_\alpha = E_\beta - i\epsilon$, times a factor $-2i\pi$. That is, in the limit $\epsilon \to 0+$, for $t \to +\infty$ the integral over α in (3.1.21) has the asymptotic behavior

$$\mathcal{I}^{\rm in}_\beta \longrightarrow -2i\pi e^{-iE_\beta t}\int d\alpha\,\delta(E_\alpha - E_\beta)g(\alpha)T^{\rm in}_{\beta\alpha},$$

and hence, for $t \to +\infty$,

$$|\Psi_g(t)\rangle_{\rm in} \longrightarrow \int d\beta\, e^{-iE_\beta t}|\beta\rangle\left[g(\beta) - 2i\pi\int d\alpha\,\delta(E_\alpha - E_\beta)g(\alpha)T^{\rm in}_{\beta\alpha}\right].$$

But expanding (3.1.19) for $|\Psi_g\rangle_{\rm in}$ in a complete set of 'out' states gives

$$|\Psi_g(t)\rangle_{\rm in} = \int d\alpha\, e^{-iE_\alpha t}g(\alpha)\int d\beta\,|\beta\rangle_{\rm out}S_{\beta\alpha}.$$

Since $S_{\beta\alpha}$ contains a factor $\delta(E_\beta - E_\alpha)$, this may be rewritten

$$|\Psi_g(t)\rangle_{\rm in} = \int d\beta\,|\beta\rangle_{\rm out}e^{-iE_\beta t}\int d\alpha\, g(\alpha)S_{\beta\alpha},$$

and, using the defining property (3.1.12) for 'out' states, this has the asymptotic behavior for $t \to +\infty$

$$|\Psi_g(t)\rangle_{\rm in} \longrightarrow \int d\beta\,|\beta\rangle e^{-iE_\beta t}\int d\alpha\, g(\alpha)S_{\beta\alpha}.$$

Comparing this with our previous result, we find

$$\int d\alpha\, g(\alpha) S_{\beta\alpha} = g(\beta) - 2i\pi \int d\alpha\, \delta(E_\alpha - E_\beta) g(\alpha) T^{\text{in}}_{\beta\alpha},$$

or in other words

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2i\pi\delta(E_\alpha - E_\beta) T^{\text{in}}_{\beta\alpha}. \tag{3.2.7}$$

This suggests a simple approximation for the S-matrix: for a weak interaction H_{int}, we can neglect the difference between 'in' and free-particle states in (3.1.18), in which case Eq. (3.2.7) gives

$$S_{\beta\alpha} \simeq \delta(\beta - \alpha) - 2i\pi\delta(E_\alpha - E_\beta)\langle\beta|H_{\text{int}}|\alpha\rangle. \tag{3.2.8}$$

This is known as the *Born approximation* [3]. Higher-order terms are discussed in Section 3.5.

* * *

We can use the Lippmann–Schwinger equations (3.1.16) for the 'in' and 'out' states to give a proof [4] of the orthonormality of these states and the unitarity of the S-matrix, as well as Eq. (3.2.7), without having to deal with limits at $t \to \mp\infty$. First, by using (3.1.16) on either the left- or right-hand side of the matrix element ${}_{\text{in/out}}\langle\beta|H_{\text{int}}|\alpha\rangle_{\text{in/out}}$ and equating the results, we find that

$$\begin{aligned} &{}_{\text{in/out}}\big\langle\beta \mid H_{\text{int}} \mid \alpha\big\rangle + {}_{\text{in/out}}\big\langle\beta \mid H_{\text{int}}(E_\alpha - H_0 \pm i\epsilon)^{-1} H_{\text{int}} \mid \alpha\big\rangle_{\text{in/out}} \\ &\quad = \big\langle\beta \mid H_{\text{int}} \mid \alpha\big\rangle_{\text{in/out}} + {}_{\text{in/out}}\big\langle\beta \mid H_{\text{int}}(E_\beta - H_0 \mp i\epsilon)^{-1} H_{\text{int}} \mid \alpha\big\rangle_{\text{in/out}}. \end{aligned}$$

Summing over a complete set $|\gamma\rangle$ of intermediate states, this gives the
equation:

$$\begin{aligned} T^{\text{in/out}\,*}_{\alpha\beta} - T^{\text{in/out}}_{\beta\alpha} = &-\int d\gamma\, T^{\text{in/out}\,*}_{\gamma\beta} T^{\text{in/out}}_{\gamma\alpha} \\ &\times \left([E_\alpha - E_\gamma \pm i\epsilon]^{-1} - [E_\beta - E_\gamma \mp i\epsilon]^{-1}\right). \end{aligned} \tag{3.2.9}$$

To prove the orthonormality of the 'in' and 'out' states, divide Eq. (3.2.9) by $E_\alpha - E_\beta \pm 2i\epsilon$. This gives

$$\left(\frac{T^{\text{in/out}}_{\alpha\beta}}{E_\beta - E_\alpha \mp 2i\epsilon}\right)^* + \frac{T^{\text{in/out}}_{\beta\alpha}}{E_\alpha - E_\beta \pm 2i\epsilon} = -\int d\gamma \left(\frac{T^{\text{in/out}}_{\gamma\beta}}{E_\beta - E_\alpha \mp i\epsilon}\right)^* \frac{T^{\text{in/out}}_{\gamma\alpha}}{E_\alpha - E_\beta \pm i\epsilon}.$$

The 2ϵs in the denominators on the left-hand side can be replaced with ϵs, since the only important thing is that these are positive infinitesimals. We see then that $\delta(\beta - \alpha) + T^{\text{in/out}}_{\beta\alpha}/(E_\beta - E_\alpha \pm i\epsilon)$ is unitary. With (3.1.17), this is just the statement that the in and out states form two orthonormal sets of state-vectors. The unitarity of the S-matrix can be proved in a similar fashion by multiplying (3.2.9) with $\delta(E_\beta - E_\alpha)$ instead of $(E_\alpha - E_\beta \pm 2i\epsilon)^{-1}$.

3.3 Symmetries of the S-Matrix

In this section we will consider both what is meant by the invariance of the S-matrix under various symmetries, and what are the conditions on the Hamiltonian that will ensure such invariance properties.

Lorentz Invariance

For any proper orthochronous Lorentz transformation $x \to \Lambda x + a$, we may define a unitary operator $U(\Lambda, a)$ by specifying that it acts as in Eq. (3.1.1) on either the 'in' or the 'out' states. When we say that a theory is Lorentz-invariant, we mean that the same operator $U(\Lambda, a)$ acts as in (3.1.1) on both 'in' and 'out' states. Since the operator $U(\Lambda, a)$ is unitary, we may write

$$S_{\beta\alpha} = {}_{\text{out}}\langle \beta \mid \alpha \rangle_{\text{in}} = {}_{\text{out}}\big\langle U(\Lambda, a)\beta \,\big|\, U(\Lambda, a)\alpha \big\rangle_{\text{in}},$$

so using (3.1.1), we obtain the Lorentz invariance (actually, covariance) property of the S-matrix: for arbitrary Lorentz transformations $\Lambda^\mu{}_\nu$ and translations a^μ,

$$\begin{aligned}
&S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\ldots,\,p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots} \\
&= \exp\big(ia_\mu(p_1^\mu + p_2^\mu + \cdots - p_1'^\mu - p_2'^\mu - \cdots)\big) \\
&\quad \times \sqrt{\frac{(\Lambda p_1)^0(\Lambda p_2)^0 \cdots (\Lambda p_1')^0(\Lambda p_2')^0 \cdots}{p_1^0 p_2^0 \cdots p_1'^0 p_2'^0 \cdots}} \\
&\quad \times \sum_{\bar\sigma_1\bar\sigma_2\ldots} D^{(j_1)}_{\bar\sigma_1\sigma_1}(W(\Lambda, p_1))\, D^{(j_2)}_{\bar\sigma_2\sigma_2}(W(\Lambda, p_2)) \cdots \\
&\quad \times \sum_{\bar\sigma_1'\bar\sigma_2'\ldots} D^{(j_1')*}_{\bar\sigma_1'\sigma_1'}\big(W(\Lambda, p_1')\big)\, D^{(j_2')*}_{\bar\sigma_2'\sigma_2'}\big(W(\Lambda, p_2')\big) \cdots \\
&\quad \times S_{\Lambda p_1'\bar\sigma_1'n_1';\Lambda p_2'\bar\sigma_2'n_2';\ldots,\,\Lambda p_1\bar\sigma_1n_1;\Lambda p_2\bar\sigma_2n_2;\ldots}.
\end{aligned} \tag{3.3.1}$$

(Primes are used to distinguish final from initial particles; bars are used to distinguish summation variables.) In particular, since the left-hand side is independent of a^μ, so must be the right-hand side, and so the S-matrix vanishes unless the four-momentum is conserved. We can therefore write the part of the S-matrix that represents actual interactions among the particles in the form:

$$S_{\beta\alpha} - \delta(\beta - \alpha) = -2\pi i M_{\beta\alpha}\delta^4(p_\beta - p_\alpha). \tag{3.3.2}$$

(However, as we will see in the next chapter, the amplitude $M_{\beta\alpha}$ itself contains terms that involve further delta function factors.)

Eq. (3.3.1) should be regarded as a definition of what we mean by the Lorentz invariance of the S-matrix, rather than a theorem, because it is only for certain special choices of Hamiltonian that there exists a unitary operator that acts as in (3.1.1) on both 'in' and 'out' states. We need to formulate conditions on the Hamiltonian that would ensure the Lorentz invariance of the S-matrix. For this purpose, it will be convenient to work with the operator S defined by Eq. (3.2.4):

$$S_{\beta\alpha} = \langle \beta | S | \alpha \rangle_0.$$

As we have defined the free-particle states in Chapter 2, they furnish a representation of the inhomogeneous Lorentz group, so we can always define a unitary operator $U_0(\Lambda, a)$ that induces the transformation (3.1.1) on these states:

$$
\begin{aligned}
&U_0(\Lambda, a)|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_0 \\
&= \exp(-ia_\mu(p_1^\mu + p_2^\mu + \cdots))\sqrt{\frac{(\Lambda p_1)^0(\Lambda p_2)^0\cdots}{p_1^0 p_2^0\cdots}} \\
&\quad\times \sum_{\bar\sigma_1\bar\sigma_2\ldots} D^{(j_1)}_{\bar\sigma_1\sigma_1}(W(\Lambda,p_1))\, D^{(j_2)}_{\bar\sigma_2\sigma_2}(W(\Lambda,p_2))\cdots|\Lambda p_1, \bar\sigma_1, n_1; \Lambda p_2, \bar\sigma_2, n_2; \ldots\rangle_0.
\end{aligned}
$$

Eq. (3.3.1) will thus hold if this unitary operator commutes with the S-operator:

$$U_0(\Lambda, a)^{-1} S\, U_0(\Lambda, a) = S.$$

This condition can also be expressed in terms of infinitesimal Lorentz transformations. Just as in Section 2.4, there will exist a set of Hermitian operators, a momentum $\mathbf{P}_0$, an angular momentum $\mathbf{J}_0$, and a boost generator $\mathbf{K}_0$, that together with H_0 generate the infinitesimal version of inhomogeneous Lorentz transformations when acting on free-particle states. Eq. (3.3.1) is equivalent to the statement that the S-matrix is unaffected by such transformations, or in other words, that the S-operator commutes with these generators:

$$[H_0, S] = [\mathbf{P}_0, S] = [\mathbf{J}_0, S] = [\mathbf{K}_0, S] = 0. \tag{3.3.3}$$

Because the operators H_0, $\mathbf{P}_0$, $\mathbf{J}_0$, and $\mathbf{K}_0$ generate infinitesimal inhomogeneous Lorentz transformations on the free states, they automatically satisfy the commutation relations (2.4.18)–(2.4.24):

$$[J_0^i, J_0^j] = i\epsilon_{ijk} J_0^k, \tag{3.3.4}$$
$$[J_0^i, K_0^j] = i\epsilon_{ijk} K_0^k, \tag{3.3.5}$$
$$[K_0^i, K_0^j] = -i\epsilon_{ijk} J_0^k, \tag{3.3.6}$$
$$[J_0^i, P_0^j] = i\epsilon_{ijk} P_0^k, \tag{3.3.7}$$
$$[K_0^i, P_0^j] = iH_0\delta_{ij}, \tag{3.3.8}$$
$$[J_0^i, H_0] = [P_0^i, H_0] = [P_0^i, P_0^j] = 0, \tag{3.3.9}$$
$$[K_0^i, H_0] = iP_0^i. \tag{3.3.10}$$

where i, j, k, etc. run over the values 1, 2, and 3, and ϵ_{ijk} is the totally antisymmetric quantity with $\epsilon_{123} = +1$.

In the same way, we may define a set of 'exact generators,' operators $\mathbf{P}$, $\mathbf{J}$, $\mathbf{K}$ that together with the full Hamiltonian H generate the transformations (3.1.1) on, say, the 'in' states. (As already mentioned, what is not obvious is that the same operators generate the same transformations on the 'out' states.) The group structure tells us that these exact generators satisfy the same commutation relations:

$$[J^i, J^j] = i\epsilon_{ijk}J^k, \tag{3.3.11}$$
$$[J^i, K^j] = i\epsilon_{ijk}K^k, \tag{3.3.12}$$
$$[K^i, K^j] = -i\epsilon_{ijk}J^k, \tag{3.3.13}$$
$$[J^i, P^j] = i\epsilon_{ijk}P^k, \tag{3.3.14}$$
$$[K^i, P^j] = iH\delta_{ij}, \tag{3.3.15}$$
$$[J^i, H] = [P^i, H] = [P^i, P^j] = 0, \tag{3.3.16}$$
$$[K^i, H] = iP^i. \tag{3.3.17}$$

In virtually all known field theories, the effect of interactions is to add an interaction term H_{int} to the Hamiltonian, while leaving the momentum and angular momentum unchanged:

$$H = H_0 + H_{\text{int}}, \qquad \mathbf{P} = \mathbf{P}_0, \qquad \mathbf{J} = \mathbf{J}_0. \tag{3.3.18}$$

(The only known exceptions are theories with topologically twisted fields, such as those with magnetic monopoles, where the angular momentum of states depends on the interactions.) Eq. (3.3.18) implies that the commutation relations (3.3.11), (3.3.14), and (3.3.16) are satisfied provided that the interaction commutes with the free-particle momentum and angular-momentum operators

$$[H_{\text{int}}, \mathbf{P}_0] = [H_{\text{int}}, \mathbf{J}_0] = 0. \tag{3.3.19}$$

It is easy to see from the Lippmann–Schwinger equation (3.1.16) or equivalently from (3.1.13) that the operators that generate translations and rotations when acting on the 'in' (and 'out') states are indeed simply $\mathbf{P}_0$ and $\mathbf{J}_0$. Also we easily see that $\mathbf{P}_0$ and $\mathbf{J}_0$ commute with the operator $U(t, t_0)$ defined by Eq. (3.2.6), and hence with the S-operator $U(\infty, -\infty)$. Further, we already know that the S-operator commutes with H_0, because there are energy-conservation delta functions in both terms in (3.2.7). This leaves just the boost generator $\mathbf{K}_0$ which we need to show commutes with the S-operator.

On the other hand, it is *not* possible to set the boost generator $\mathbf{K}$ equal to its free-particle counterpart $\mathbf{K}_0$, because then Eqs. (3.3.15) and (3.3.8) would give $H = H_0$, which is certainly not true in the presence of interactions. Thus when we add an interaction H_{int} to H_0, we must also add a correction $\mathbf{W}$ to the boost generator:

$$\mathbf{K} = \mathbf{K}_0 + \mathbf{W}. \tag{3.3.20}$$

Of the remaining commutation relations, let us concentrate on Eq. (3.3.17), which may now be put in the form

$$[\mathbf{K}_0, H_{\text{int}}] = -[\mathbf{W}, H]. \tag{3.3.21}$$

By itself, the condition (3.3.21) is empty, because for any H_{int} we could always define $\mathbf{W}$ by giving its matrix elements between H-eigenstates $|\alpha\rangle$ and $|\beta\rangle$ as $-\langle\beta|[\mathbf{K}_0, H_{\text{int}}]|\alpha\rangle/(E_\beta - E_\alpha)$. Recall that the crucial point in the Lorentz invariance of a theory is not that there should exist a set of exact generators satisfying Eqs. (3.3.11)–(3.3.17), but rather that these operators should act the same way on 'in' and 'out' states; merely finding an operator $\mathbf{K}$ that satisfies Eq. (3.3.21) is not enough. Eq. (3.3.21) does become significant if we add the requirement that matrix elements of $\mathbf{W}$ should be smooth functions of the energies, and in particular should not have singularities of the form $(E_\beta - E_\alpha)^{-1}$. We shall now show that Eq. (3.3.21), together with an appropriate smoothness condition on $\mathbf{W}$, does imply the remaining Lorentz invariance condition $[\mathbf{K}_0, S] = 0$.

To prove this, let us consider the commutator of $\mathbf{K}_0$ with the operator $U(t, t_0)$ for finite t and t_0. Using Eq. (3.3.10) and the fact that $\mathbf{P}_0$ commutes with H_0 yields:

$$[\mathbf{K}_0, \exp(iH_0 t)] = -t\mathbf{P}_0 \exp(iH_0 t)$$

while Eq. (3.3.21) (which is equivalent to Eq. (3.3.17)) yields

$$[\mathbf{K}, \exp(iHt)] = -t\mathbf{P}\exp(iHt) = -t\mathbf{P}_0 \exp(iHt).$$

The momentum operators then cancel in the commutator of $\mathbf{K}_0$ with U, and we find:

$$[\mathbf{K}_0, U(\tau, \tau_0)] = -\mathbf{W}(\tau)U(\tau, \tau_0) + U(\tau, \tau_0)\mathbf{W}(\tau_0), \tag{3.3.22}$$

where

$$\mathbf{W}(t) = \exp(iH_0 t)\mathbf{W}\exp(-iH_0 t). \tag{3.3.23}$$

If the matrix elements of $\mathbf{W}$ between H_0-eigenstates are sufficiently smooth functions of energy, then matrix elements of $\mathbf{W}(t)$ between smooth superpositions of energy eigenstates vanish for $t \to \pm\infty$, so Eq. (3.3.22) gives in effect:

$$0 = [\mathbf{K}_0, U(\infty, -\infty)] = [\mathbf{K}_0, S], \tag{3.3.24}$$

as was to be shown. This is the essential result: Eq. (3.3.21) together with the smoothness condition on matrix elements of $\mathbf{W}$ that ensures that $\mathbf{W}(t)$ vanishes for $t \to \pm\infty$ provides a sufficient condition for the Lorentz invariance of the S-matrix. This smoothness condition is a natural one, because it is much like the condition on matrix elements of H_{int} that is needed to make $H_I(t)$ vanish for $t \to \pm\infty$, as required in order to justify the very idea of an S-matrix.

We can also use Eq. (3.3.22) with $\tau = 0$ and $\tau_0 = \mp\infty$ to show that

$$\mathbf{K}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{K}_0, \tag{3.3.25}$$

where $\Omega(\mp\infty)$ is according to (3.1.13) the operator that converts a free-particle state $|\alpha\rangle_0$ into the corresponding 'in' or 'out' state. Also, it follows trivially from Eqs. (3.3.18) and (3.3.19) that the same is true for the momentum and angular momentum:

$$\mathbf{P}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{P}_0, \tag{3.3.26}$$

$$\mathbf{J}\Omega(\mp\infty) = \Omega(\mp\infty)\mathbf{J}_0. \tag{3.3.27}$$

Finally, since all free and asymptotic states are eigenstates of H_0 and H respectively with the same eigenvalue E_α, we have

$$H\Omega(\mp\infty) = \Omega(\mp\infty)H_0. \tag{3.3.28}$$

Eqs. (3.3.25)–(3.3.28) show that with our assumptions, 'in' and 'out' states do transform under inhomogeneous Lorentz transformations just like the free-particle states. Also, since these are similarity transformations, we now see that the exact generators $\mathbf{K}$, $\mathbf{P}$, $\mathbf{J}$, and H satisfy the same commutation relations as $\mathbf{K}_0$, $\mathbf{P}_0$, $\mathbf{J}_0$, and H_0. This is why it turned out to be unnecessary in proving the Lorentz invariance of the S-matrix to use the other commutation relations (3.3.12), (3.3.13), and (3.3.15) that involve $\mathbf{K}$.

Internal Symmetries

There are various symmetries, like the symmetry in nuclear physics under interchange of neutrons and protons, or the 'charge-conjugation' symmetry between particles and antiparticles, that have nothing directly to do with Lorentz invariance, and further appear the same in all inertial frames. Such a symmetry transformation T acts on the Hilbert space of physical states as a unitary operator $U(T)$, that induces linear transformations on the indices labelling particle species

$$\begin{aligned} &U(T)|p_1,\sigma_1,n_1;p_2,\sigma_2,n_2;\ldots\rangle_{\text{in/out}} \\ &\quad = \sum_{\bar{n}_1\bar{n}_2\ldots} \mathcal{D}_{\bar{n}_1 n_1}(T)\mathcal{D}_{\bar{n}_2 n_2}(T)\cdots|p_1,\sigma_1,\bar{n}_1;p_2,\sigma_2,\bar{n}_2;\ldots\rangle_{\text{in/out}}. \end{aligned} \tag{3.3.29}$$

In accordance with the general discussion in Chapter 2, the $U(T)$ must satisfy the group multiplication rule

$$U(\bar{T})U(T) = U(\bar{T}T), \tag{3.3.30}$$

where $\bar{T}T$ is the transformation obtained by first performing the transformation T, then some other transformation $\bar{T}$. Acting on Eq. (3.3.29) with $U(\bar{T})$, we see that the matrices $\mathcal{D}$ satisfy the same rule

$$\mathcal{D}(\bar{T})\mathcal{D}(T) = \mathcal{D}(\bar{T}T). \tag{3.3.31}$$

Also, taking the scalar product of the states obtained by acting with $U(T)$ on two different 'in' states or two different 'out' states, and using the normalization condition (3.1.2), we see that $\mathcal{D}(T)$ must be unitary

$$\mathcal{D}^\dagger(T) = \mathcal{D}^{-1}(T). \tag{3.3.32}$$

Finally, taking the scalar product of the states obtained by acting with $U(T)$ on one 'out' state and one 'in' state shows that $\mathcal{D}$ commutes with the S-matrix, in the sense that

$$\begin{aligned} &\sum_{\bar{N}_1\bar{N}_2\ldots}\sum_{\bar{N}'_1\bar{N}'_2\ldots} \mathcal{D}^*_{\bar{N}'_1 n'_1}(T)\mathcal{D}^*_{\bar{N}'_2 n'_2}(T)\cdots\mathcal{D}_{\bar{N}_1 n_1}(T)\mathcal{D}_{\bar{N}_2 n_2}(T)\cdots \\ &\qquad \times S_{p'_1\sigma'_1\bar{N}'_1;p'_2\sigma'_2\bar{N}'_2;\ldots,\,p_1\sigma_1\bar{N}_1;p_2\sigma_2\bar{N}_2;\ldots} \\ &\quad = S_{p'_1\sigma'_1 n'_1;p'_2\sigma'_2 n'_2;\ldots,\,p_1\sigma_1 n_1;p_2\sigma_2 n_2;\ldots}. \end{aligned} \tag{3.3.33}$$

Again, this is a definition of what we mean by a theory being invariant under the internal symmetry T, because to derive Eq. (3.3.33) we still need to show that the *same* unitary operator $U(T)$ will induce the transformation (3.3.29) on both 'in' and 'out' states.

This will be the case if there is an 'unperturbed' transformation operator $U_0(T)$ that induces these transformations on free-particle states,

$$U_0(T)|p_1,\sigma_1,n_1;p_2,\sigma_2,n_2;\ldots\rangle_0$$
$$=\sum_{\bar N_1\bar N_2\ldots}\mathcal{D}_{\bar N_1 n_1}(T)\mathcal{D}_{\bar N_2 n_2}(T)\cdots|p_1,\sigma_1,\bar N_1;p_2,\sigma_2,\bar N_2;\ldots\rangle_0 \tag{3.3.34}$$

and that commutes with both the free-particle and interaction parts of the Hamiltonian

$$U_0^{-1}(T)H_0U_0(T)=H_0, \tag{3.3.35}$$
$$U_0^{-1}(T)H_{\rm int}U_0(T)=H_{\rm int}. \tag{3.3.36}$$

From either the Lippmann–Schwinger equation (3.1.17) or from (3.1.13), we see that the operator $U_0(T)$ will induce the transformations (3.3.29) on 'in' and 'out' states as well as free-particle states, so that we can derive Eq. (3.3.29) taking $U(T)$ as $U_0(T)$.

A special case of great physical importance is that of a one-parameter Lie group, where T is a function of a single parameter θ, with

$$T(\bar\theta)T(\theta)=T(\bar\theta+\theta). \tag{3.3.37}$$

As shown in Section 2.2, in this case the corresponding Hilbert-space operators must take the form

$$U(T(\theta))=\exp(iQ\theta) \tag{3.3.38}$$

with Q a Hermitian operator. Likewise the matrices $\mathcal{D}(T)$ take the form

$$\mathcal{D}_{n'n}(T(\theta))=\delta_{n'n}\exp(iq_n\theta), \tag{3.3.39}$$

where q_n are a set of real species-dependent numbers. Here Eq. (3.3.33) simply tells us that the qs are conserved: $S_{\beta\alpha}$ vanishes unless

$$q_{n'_1}+q_{n'_2}+\cdots=q_{n_1}+q_{n_2}+\cdots\,. \tag{3.3.40}$$

The classic example of such a conservation law is that of conservation of electric charge. Also, all known processes conserve baryon number (the number of baryons, such as protons, neutrons, and hyperons, minus the number of their antiparticles) and lepton number (the number of leptons, such as electrons, muons, τ particles, and neutrinos, minus the number of their antiparticles) but as we shall see in Volume II, these conservation laws are believed to be only very good approximations. There are other conservation laws of this type that are definitely only approximate, such as the conservation of the quantity known as strangeness, which was introduced to explain the relatively long life of a class of particles discovered by Rochester and Butler [5] in cosmic rays in 1947. For instance, the mesons now called K^+ and K^0[4] are assigned strangeness $+1$ and the hyperons Λ^0, Σ^+, Σ^0, Σ^- are assigned strangeness -1, while the more familiar protons, neutrons, and π mesons (or pions) are

[4]Superscripts denote electric charges in units of the absolute value of the electronic charge. A 'hyperon' is any particle carrying non-zero strangeness and unit baryon number.

taken to have strangeness zero. The conservation of strangeness in strong interactions explains why strange particles are always produced in association with one another, as in reactions like $\pi^+ + n \to K^+ + \Lambda^0$, while the relatively slow decays of strange particles into non-strange ones such as $\Lambda^0 \to p + \pi^-$ and $K^+ \to \pi^+ + \pi^0$ show that the interactions that do not conserve strangeness are very weak.

The classic example of a 'non-Abelian' symmetry whose generators do not commute with one another is isotopic spin symmetry, which was suggested [6] in 1937 on the basis of an experiment [7] that showed the existence of a strong proton–proton force similar to that between protons and neutrons. Mathematically, the group is $SU(2)$, like the covering group of the group of three-dimensional rotations; its generators are denoted t_i, with $i = 1, 2, 3$, and satisfy a commutation relation like (2.4.18):

$$[t_i, t_j] = i\epsilon_{ijk} t_k.$$

To the extent that isotopic spin symmetry is respected, it requires particles to form degenerate multiplets labelled with an integer or half-integer T and with $2T + 1$ components distinguished by their t_3 values, just like the degenerate spin multiplets required by rotational invariance. These include the nucleons p and n with $T = \frac{1}{2}$ and $t_3 = \frac{1}{2}, -\frac{1}{2}$; the pions π^+, π^0, and π^- with $T = 1$ and $t_3 = +1, 0, -1$; and the Λ^0 hyperon with $T = 0$ and $t_3 = 0$. These examples illustrate the relation between electric charge Q, the third component of isotopic spin t_3, the baryon number B, and the strangeness S:

$$Q = t_3 + (B + S)/2.$$

This relation was originally inferred from observed selection rules, but it was interpreted [8] by Gell-Mann and Ne'eman in 1960 to be a consequence of the embedding of both the isospin $\mathbf{T}$ and the 'hypercharge' $Y \equiv B + S$ in the Lie algebra of a larger but more badly broken non-Abelian internal symmetry, based on the non-Abelian group $SU(3)$. As we will see in Volume II, today both isospin and $SU(3)$ symmetry are understood as incidental consequences of the small masses of the two or three lightest quarks in the modern theory of strong interactions, quantum chromodynamics.

The implications of isotopic spin symmetry for reactions among strongly interacting particles can be worked out by the same familiar methods that were invented for deriving the implications of rotational invariance. In particular, for a two-body reaction $A+B \to C+D$, Eq. (3.3.33) requires that the S-matrix may be put in the form (suppressing all but isospin labels):

$$S_{t_{C3}t_{D3},t_{A3}t_{B3}} = \sum_{T,t_3} C_{T_C T_D}(T t_3; t_{C3} t_{D3}) C_{T_A T_B}(T t_3; t_{A3} t_{B3}) S_T,$$

where $C_{j_1 j_2}(j\sigma; \sigma_1\sigma_2)$ is the usual Clebsch–Gordan coefficient [9] for forming a spin j with three-component σ from spins j_1 and j_2 with three-components σ_1 and σ_2, respectively; and S_T is a 'reduced' S-matrix depending on T and on all the suppressed momentum and spin variables, but not on the isospin three-components $t_{A3}, t_{B3}, t_{C3}, t_{D3}$. Of course, this, like all of the consequences of isotopic spin invariance, is only approximate, because this symmetry is not respected by electromagnetic (and other) interactions, as shown for instance by the fact that different members of the same isospin multiplet like p and n have different electric charges and slightly different masses.

Parity

To the extent that the symmetry under the transformation $\mathbf{x} \to -\mathbf{x}$ is really valid, there must exist a unitary operator P under which both 'in' and 'out' states transform as a direct product of single-particle states:

$$P|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} = \eta_{n_1}\eta_{n_2}\cdots|\mathcal{P}p_1, \sigma_1, n_1; \mathcal{P}p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}}. \quad (3.3.41)$$

where η_n is the intrinsic parity of particles of species n, and $\mathcal{P}$ reverses the space components of p^μ. (This is for massive particles; the modification for massless particles is obvious.) The parity conservation condition for the S-matrix is then:

$$\begin{aligned} &S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\ldots,\, p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots} \\ &\quad = \eta_{n_1'}^*\eta_{n_2'}^*\cdots\eta_{n_1}\eta_{n_2}\cdots S_{\mathcal{P}p_1'\sigma_1'n_1';\mathcal{P}p_2'\sigma_2'n_2';\ldots,\, \mathcal{P}p_1\sigma_1n_1;\mathcal{P}p_2\sigma_2n_2;\ldots}. \end{aligned} \quad (3.3.42)$$

Just as for internal symmetries, an operator P satisfying Eq. (3.3.41) will actually exist if the operator P_0 which is defined to act this way on free-particle states commutes with H_{int} as well as H_0.

The phases η_n may be inferred either from dynamical models or from experiment, but neither can provide a unique determination of the ηs. This is because we are always free to redefine P by combining it with any conserved internal symmetry operator. For instance, if P is conserved, then so is

$$P' \equiv P\exp(i\alpha B + i\beta L + i\gamma Q),$$

where B, L, and Q are respectively baryon number, lepton number, and electric charge, and α, β, and γ are arbitrary real phases; hence either P or P' could be called the parity operator. The neutron, proton, and electron have different combinations of values for B, L, and Q, so by judicious choice of the phases α, β, and γ we can define the intrinsic parities of all three particles to be $+1$. However, once we have done this the intrinsic parities of other particles like the charged pion (which can be emitted in a transition $n \to p + \pi^-$) are no longer arbitrary. Also, the intrinsic parity of any particle like the neutral pion π^0 which carries no conserved quantum numbers is always meaningful.

The foregoing remarks help to clarify the question of whether intrinsic parities must always have the values ± 1. It is easy to say that space inversion $\mathcal{P}$ has the group multiplication law $\mathcal{P}^2 = \mathbf{1}$; however, the parity operator that is conserved may not be this one, but rather may differ from it by a phase transformation of some sort. In any case, whether or not $P^2 = \mathbf{1}$, the operator P^2 behaves just like an internal symmetry transformation:

$$P^2|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}} = \eta_{n_1}^2\eta_{n_2}^2\cdots|p_1, \sigma_1, n_1; p_2, \sigma_2, n_2; \ldots\rangle_{\text{in/out}}.$$

If this internal symmetry is part of a continuous symmetry group of phase transformations, such as the group of multiplication by the phases $\exp(i\alpha B + i\beta L + i\gamma Q)$ with arbitrary values of α, β, and γ, then its inverse square root must also be a member of this group, say I_P, with $I_P^2 P^2 = \mathbf{1}$ and $[I_P, P] = 0$. (For instance, if $P^2 = \exp(i\alpha B + \cdots)$, then take $I_P = \exp(-\frac{1}{2}i\alpha B + \cdots)$.) We can then define a new parity operator $P' \equiv PI_P$ with $P'^2 = \mathbf{1}$. This is conserved to the same extent as P, so there is no reason why we should not call this *the* parity operator, in which case the intrinsic parities can only take the values ± 1.

The only sort of theory in which it is not necessarily possible to define parity so that all intrinsic parities have the values ± 1 is one in which there is some discrete internal symmetry which is not a member of any continuous symmetry group of phase transformations [10]. For instance, it is a consequence of angular-momentum conservation that the total number F of all particles of half-integer spin can only change by even numbers, so the internal symmetry operator $(-1)^F$ is conserved. All known particles of half-integer spin have odd values of the sum $B + L$ of baryon number and lepton number, so as far as we know, $(-1)^F = (-1)^{B+L}$. If this is true, then $(-1)^F$ is part of a continuous symmetry group, consisting of the operators $\exp(i\alpha(B + L))$ with arbitrary real α, and has an inverse square root $\exp(-i\pi(B + L)/2)$. In this case, if $P^2 = (-1)^F$ then P can be redefined so that all intrinsic parities are ± 1. However, if there were to be discovered a particle of half-integer spin and an even value of $B + L$ (such as a so-called Majorana neutrino, with $j = \frac{1}{2}$ and $B + L = 0$), then it would be possible to have $P^2 = (-1)^F$ without our being able to redefine the parity operator itself to have eigenvalues ± 1. In this case, of course, we would have $P^4 = \mathbf{1}$, so all particles would have intrinsic parities either ± 1 or (like the Majorana neutrino) $\pm i$.

It follows from Eq. (3.3.42) that if the product of intrinsic parities in the final state is equal to the product of intrinsic parities in the initial state, or equal to minus this product, then the S-matrix must be respectively even or odd overall in the three-momenta. For instance, it was observed [11] in 1951 that a pion can be absorbed by a deuteron from the $\ell = 0$ ground state of the $\pi^- d$ atom, in the reaction $\pi^- + d \to n + n$. (As discussed in Section 3.7, the orbital angular-momentum quantum number ℓ can be used in relativistic physics in the same way as in non-relativistic wave mechanics.) The initial state has total angular-momentum $j = 1$ (the pion and deuteron having spins zero and one, respectively), so the final state must have orbital angular-momentum $\ell = 1$ and total neutron spin $s = 1$. (The other possibilities $\ell = 1$, $s = 0$; $\ell = 0$, $s = 1$; and $\ell = 2$, $s = 1$, which are allowed by angular-momentum conservation, are forbidden by the requirement that the final state be antisymmetric in the two neutrons.) Because the final state has $\ell = 1$, the matrix element is odd under reversal of the direction of all three-momenta, so we can conclude that the intrinsic parities of the particles in this reaction must be related by:

$$\eta_d \eta_{\pi^-} = -\eta_n^2 .$$

The deuteron is known to be a bound state of a proton and neutron with even orbital angular-momentum (chiefly $\ell = 0$), and as we have seen we can take the neutron and proton to have the same intrinsic parity, so $\eta_d = \eta_n^2$, and we can conclude that $\eta_{\pi^-} = -1$; that is, the negative pion is a *pseudoscalar* particle. The π^+ and π^0 have also been found to have negative parity, as would be expected from the symmetry (isospin invariance) among these three particles.

The negative parity of the pion has some striking consequences. A spin zero particle that decays into three pions must have intrinsic parity $\eta_\pi^3 = -1$, because in the Lorentz frame in which the decaying particle is at rest, rotational invariance only allows the matrix element to depend on scalar products of the pion momenta with each other, all of which are even under reversal of all momenta. (The triple scalar product $\mathbf{p}_1 \cdot (\mathbf{p}_2 \times \mathbf{p}_3)$ formed from the three pion momenta vanishes because $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 = 0$.) For the same reason, a spin zero particle that decays into two pions must have intrinsic parity $\eta_\pi^2 = +1$. In particular, among the strange particles discovered in the late 1940s there seemed to be two different particles of zero spin (inferred from the angular distribution of their decay products): one, the τ, was identified by its decay into three pions, and hence was assigned a parity -1, while the other, the θ, was identified by its decay into two pions and was assigned a parity $+1$. The trouble with all this was that as the τ and θ were studied in greater detail, they seemed increasingly to have identical masses and lifetimes. After many suggested solutions of this puzzle, Lee and Yang in 1956 finally cut the Gordian knot, and proposed that the τ and θ are the same particle, (now known as the $K^\pm$) and that parity is simply not conserved in the weak interactions that lead to its decay [12].

As we shall see in detail in the next section of this chapter, the rate for a physical process $\alpha \to \beta$ (with $\alpha \neq \beta$) is proportional to $|S_{\beta\alpha}|^2$, with proportionality factors that are invariant under reversal of all three-momenta. As long as the states α and β contain definite numbers of particles of each type, the phase factors in Eq. (3.3.42) have no effect on $|S_{\beta\alpha}|^2$, so Eq. (3.3.42) would imply that the rate for $\alpha \to \beta$ is invariant under the reversal of direction of all three-momenta. As we have seen, this is a trivial consequence of rotational invariance for the decays of a K meson into two or three pions, but it is a non-trivial restriction on rates in more complicated processes. For example, following theoretical suggestions by Lee and Yang, Wu together with a group at the National Bureau of Standards measured the angular distribution of the electron in the final state of the beta decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ with a polarized cobalt source [13]. (No attempt was made in this experiment to measure the momentum of the antineutrino or nickel nucleus.) The electrons were found to be preferentially emitted in a direction opposite to that of the spin of the decaying nucleus, which would, of course, be impossible if the decay rate were invariant under a reversal of all three-momenta. A similar result was found in the decay of a positive muon (polarized in its production in the process $\pi^+ \to \mu^+ + \nu$) into a positron, neutrino, and antineutrino [14]. In this way, it became clear that parity is indeed not conserved in the weak interactions responsible for these decays. Nevertheless, for reasons discussed in Section 12.5, parity *is* conserved in the strong and electromagnetic interactions, and therefore continues to play an important part in theoretical physics.

Time-Reversal

We saw in Section 2.6 that the time-reversal operator T acting on a one-particle state $|p, \sigma, n\rangle$ gives a state $|\mathcal{P}p, -\sigma, n\rangle$ with reversed spin and momentum, times a phase $\zeta_n(-1)^{j-\sigma}$. A multi-particle state transforms as usual as a direct product of one-particle states, except that since this is a time-reversal transformation, we expect 'in' and 'out' states to be interchanged:

$$\begin{aligned} &T|p_1,\sigma_1,n_1;p_2,\sigma_2,n_2;\ldots\rangle_{\rm in/out} \\ &\quad = \zeta_{n_1}(-1)^{j_1-\sigma_1}\zeta_{n_2}(-1)^{j_2-\sigma_2}\cdots|\mathcal{P}p_1,-\sigma_1,n_1;\mathcal{P}p_2,-\sigma_2,n_2;\ldots\rangle_{\rm out/in}. \end{aligned} \tag{3.3.43}$$

(Again, this is for massive particles, with obvious modifications being required for massless particles.) It will be convenient to abbreviate this assumption as

$$T|\alpha\rangle_{\rm in/out} = |\Theta\alpha\rangle_{\rm out/in}, \tag{3.3.44}$$

where Θ indicates a reversal of sign of three-momenta and spins as well as multiplication by the phase factors shown in Eq. (3.3.43). Because T is antiunitary, we have

$${}_{\rm out}\langle\beta\mid\alpha\rangle_{\rm in} = {}_{\rm out}\langle\Theta\alpha\mid\Theta\beta\rangle_{\rm in}, \tag{3.3.45}$$

so the time-reversal invariance condition for the S-matrix is

$$S_{\beta\alpha} = S_{\Theta\alpha,\Theta\beta} \tag{3.3.46}$$

or in more detail

$$\begin{aligned} &S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\ldots,\,p_1\sigma_1n_1;p_2\sigma_2n_2;\ldots} \\ &\quad = \zeta_{n_1'}(-1)^{j_1'-\sigma_1'}\zeta_{n_2'}(-1)^{j_2'-\sigma_2'}\cdots\zeta^*_{n_1}(-1)^{j_1-\sigma_1}\zeta^*_{n_2}(-1)^{j_2-\sigma_2}\cdots \\ &\qquad \times S_{\mathcal{P}p_1,-\sigma_1,n_1;\mathcal{P}p_2,-\sigma_2,n_2;\ldots,\,\mathcal{P}p_1',-\sigma_1',n_1';\mathcal{P}p_2',-\sigma_2',n_2';\ldots}. \end{aligned} \tag{3.3.47}$$

Note that in addition to the reversal of momenta and spins, the role of initial and final states is interchanged, as would be expected for a symmetry involving the reversal of time.

The S-matrix will satisfy this transformation rule if the operator T_0 that induces time-reversal transformations on free-particle states

$$T_0|\alpha\rangle_0 = |\Theta\alpha\rangle_0 \tag{3.3.48}$$

commutes not only with the free-particle Hamiltonian (which is automatic) but also with the interaction:

$$T_0^{-1}H_0T_0 = H_0, \tag{3.3.49}$$

$$T_0^{-1}H_{\rm int}T_0 = H_{\rm int}. \tag{3.3.50}$$

In this case we can take $T = T_0$, and use either (3.1.13) or (3.1.16) to show that time-reversal transformations do act as stated in Eq. (3.3.44). For instance, operating on the Lippmann–Schwinger equation (3.1.16) with T and using Eqs. (3.3.48)–(3.3.50), we have

$$T|\alpha\rangle_{\rm in/out} = |\Theta\alpha\rangle_0 + \int d\gamma\,[E_\alpha - H_0 \mp i\epsilon]^{-1}H_{\rm int}T|\alpha\rangle_{\rm in/out}$$

with the sign of $\pm i\epsilon$ reversed because T is antiunitary. This is just the Lippmann–Schwinger equation for $|\Theta\alpha\rangle_{\rm out/in}$, thus justifying Eq. (3.3.44). Similarly, because T is antiunitary it changes the sign of the i in the exponent of $\Omega(t)$, so that

$$T\Omega(-\infty)|\alpha\rangle_0 = \Omega(\infty)|\Theta\alpha\rangle_0,$$

again leading to Eq. (3.3.44).

In contrast with the case of parity conservation, the time-reversal invariance condition (3.3.46) does *not* in general tell us that the rate for the process $\alpha \to \beta$ is the same as for the process $\Theta\alpha \to \Theta\beta$. However, something like this is true in cases where the S-matrix takes the form

$$S_{\beta\alpha} = S^{(0)}_{\beta\alpha} + S^{(1)}_{\beta\alpha}, \tag{3.3.51}$$

where $S^{(1)}$ is small, while $S^{(0)}$ happens to have matrix element zero for some particular process of interest, though it generally has much larger matrix elements than $S^{(1)}$. (For instance, the process might be nuclear beta decay, $N \to N' + e^- + \bar{\nu}$, with $S^{(0)}$ the S-matrix produced by the strong nuclear and electromagnetic interactions alone, and $S^{(1)}$ the correction to the S-matrix produced by the weak interactions. Section 3.5 shows how the use of the 'distorted-wave Born approximation' leads to an S-matrix of the form (3.3.51) in cases of this type. In some cases $S^{(0)}$ is simply the unit operator.) To first order in $S^{(1)}$, the unitarity condition for the S-operator reads

$$\mathbf{1} = S^\dagger S = S^{(0)\dagger}S^{(0)} + S^{(0)\dagger}S^{(1)} + S^{(1)\dagger}S^{(0)}.$$

Using the zeroth-order relation $S^{(0)\dagger}S^{(0)} = \mathbf{1}$, this gives a reality condition for $S^{(1)}$:

$$S^{(1)} = -S^{(0)}S^{(1)\dagger}S^{(0)}. \tag{3.3.52}$$

If $S^{(1)}$ as well as $S^{(0)}$ satisfies the time-reversal condition (3.3.46), then this can be put in the form

$$S^{(1)}_{\beta\alpha} = -\int d\gamma \int d\gamma'\, S^{(0)}_{\beta\gamma'} S^{(1)*}_{\Theta\gamma',\Theta\gamma} S^{(0)}_{\gamma\alpha}. \tag{3.3.53}$$

Since $S^{(0)}$ is unitary, the rates for the processes $\alpha \to \beta$ and $\Theta\alpha \to \Theta\beta$ are thus the same if summed over sets $\mathcal{J}$ and $\mathcal{F}$ of final and initial states that are complete with respect to $S^{(0)}$. (By being 'complete' here is meant that if $S^{(0)}_{\alpha'\alpha}$ is non-zero, and either α or α' are in $\mathcal{J}$, then both states are in $\mathcal{J}$; and similarly for $\mathcal{F}$.) In the simplest case we have 'complete' sets $\mathcal{J}$ and $\mathcal{F}$ consisting of just one state each; that is, both the initial and the final states are eigenvectors of $S^{(0)}$ with eigenvalues $e^{2i\delta_\alpha}$ and $e^{2i\delta_\beta}$, respectively. (The δ_α and δ_β are called 'phase shifts'; they are real because $S^{(0)}$ is unitary.) In this case, Eq. (3.3.53) becomes simply:

$$S^{(1)}_{\beta\alpha} = -e^{2i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\Theta\beta,\Theta\alpha}, \tag{3.3.54}$$

and it is clear that the absolute value of the S-matrix for the process $\alpha \to \beta$ is the same as for the process $\Theta\alpha \to \Theta\beta$. This is the case for instance in nuclear beta decay (in the approximation in which we ignore the relatively weak Coulomb interaction between the electron and nucleus in the final state), because both the initial and final states are eigenstates of the strong interaction S-matrix (with $\delta_\alpha = \delta_\beta = 0$). Thus if time-reversal invariance is respected, the differential rate for a beta decay process should be unchanged if we reverse both the momenta and the spin z-components σ of all particles. This prediction was *not* contradicted in the 1956 experiments [13,14] that discovered the non-conservation of parity; for instance, time-reversal invariance is consistent with the observation that electrons from the decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ are preferentially emitted in a direction opposite to that of the Co^{60} spin. As described below, indirect evidence against time-reversal invariance did emerge in 1964, but it remains a useful approximate symmetry in weak as well as strong and electromagnetic interactions.

In some cases we can use a basis of states for which $\Theta\alpha = \alpha$ and $\Theta\beta = \beta$, for which Eq. (3.3.54) reads

$$S^{(1)}_{\beta\alpha} = -e^{2i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\beta\alpha}, \tag{3.3.55}$$

which just says that $iS^{(1)}_{\beta\alpha}$ has the phase $\delta_\alpha + \delta_\beta \mod \pi$. This is known as *Watson's theorem* [15]. The phases in Eqs. (3.3.54) or (3.3.55) may be measured in processes where there is interference between different final states. For instance, in the decay of the spin 1/2 hyperon Λ into a nucleon and a pion, the final state can only have orbital angular-momentum $\ell = 0$ or $\ell = 1$; the angular distribution of the pion relative to the Λ spin involves the interference between these states, and hence according to Watson's theorem depends on the difference $\delta_s - \delta_p$ of their phase shifts.

PT

Although the 1957 experiments on parity violation did not rule out time-reversal invariance, they did show immediately that the product PT is not conserved. If conserved this operator would have to be antiunitary for the same reasons as for T, so in processes like nuclear beta decay its consequences would take the form of relations like Eq. (3.3.54):

$$S^{(1)}_{\beta\alpha} = -e^{i(\delta_\alpha+\delta_\beta)} S^{(1)*}_{\mathcal{P}\Theta\beta,\mathcal{P}\Theta\alpha},$$

where $\mathcal{P}\Theta$ reverses the signs of all spin z-components but *not* any momenta. Neglecting the final-state Coulomb interaction, it would then follow that there could be no preference for the electron in the decay $\mathrm{Co}^{60} \to \mathrm{Ni}^{60} + e^- + \bar{\nu}$ to be emitted in the same or the opposite direction to the Co^{60} spin, in contradiction with what was observed.

C, *CP*, and *CPT*

As already mentioned, there is an internal symmetry transformation, known as charge-conjugation, which interchanges particles and antiparticles. Formally, this entails the existence of a unitary operator C, whose effect on multi-particle states is:

$$C|p_1,\sigma_1,n_1;p_2,\sigma_2,n_2;\dots\rangle_{\text{in/out}} = \xi_{n_1}\xi_{n_2}\cdots|p_1,\sigma_1,n_1^c;p_2,\sigma_2,n_2^c;\dots\rangle_{\text{in/out}}, \tag{3.3.56}$$

where n^c is the antiparticle of particle type n, and ξ_n is yet another phase. If this is true for both 'in' and 'out' states then the S-matrix satisfies the invariance conditions

$$\begin{aligned} &S_{p_1'\sigma_1'n_1';p_2'\sigma_2'n_2';\dots,\,p_1\sigma_1n_1;p_2\sigma_2n_2;\dots} \\ &\quad = \xi^*_{n_1'}\xi^*_{n_2'}\cdots\xi_{n_1}\xi_{n_2}\cdots S_{p_1'\sigma_1'n_1'^c;p_2'\sigma_2'n_2'^c;\dots,\,p_1\sigma_1n_1^c;p_2\sigma_2n_2^c;\dots}. \end{aligned} \tag{3.3.57}$$

As with other internal symmetries, the S-matrix will satisfy this condition if the operator C_0 that is defined to act as stated in Eq. (3.3.56) on free-particle states commutes with the interaction $H_{\rm int}$ as well as H_0; in this case, we take $C = C_0$.

The phases ξ_n are called charge-conjugation parities. Just as for the ordinary parities η_n, the ξ_n are in general not uniquely defined, because for any operator C that is defined to satisfy Eq. (3.3.56), we can find another such operator with different ξ_n by multiplying C with any internal symmetry phase transformation, such as $\exp(i\alpha B + i\beta L + i\gamma Q)$; the only particles whose charge-conjugation parities are individually measurable are those completely neutral particles like the photon or the neutral pion that carry no conserved quantum numbers and are their own antiparticles. In reactions involving only completely neutral particles, Eq. (3.3.57) tells us that the product of the charge-conjugation parities in the initial and final states must be equal; for instance, as we shall see the photon is required by quantum electrodynamics to have charge-conjugation parity $\eta_\gamma = -1$, so the observation of the neutral pion decay $\pi^0 \to 2\gamma$ requires that $\eta_{\pi^0} = +1$; it then follows that the process $\pi^0 \to 3\gamma$ should be forbidden, as is in fact known to be the case. For these two particles, the charge-conjugation parities are real, either $+1$ or -1. Just as for ordinary parity, this will always be the case if all internal phase transformation symmetries are members of continuous groups of phase transformations, because then we can redefine C by multiplying by the inverse square root of the internal symmetry equal to C^2, with the result that the new C satisfies $C^2 = \mathbf{1}$.

For general reactions, Eq. (3.3.57) requires that the rate for a process equals the rate for the same process with particles replaced with their corresponding antiparticles. This was not directly contradicted by the 1957 experiments on parity non-conservation (it will be a long time before anyone is able to study the beta decay of anticobalt), but these experiments showed that C is not conserved in the *theory* of weak interactions as modified by Lee and Yang [12] to take account of parity non-conservation. (As we shall see below, the observed violation of TP conservation would imply a violation of C conservation in *any* field theory of weak interactions, not just in the particular theory considered by Lee and Yang.) It is understood today that C as well as P is not conserved in the weak interactions responsible for processes like beta decay and the decay of the pion and muon, though both C and P are conserved in the strong and electromagnetic interactions.

Although the early experiments on parity non-conservation indicated that neither C nor P are conserved in the weak interactions, they left open the possibility that their product CP is universally conserved. For some years it was expected (though not with complete confidence) that CP would be found to be generally conserved. This had particularly important consequences for the properties of the neutral K mesons. In 1954 Gell-Mann and Pais [16] had pointed out that because the K^0 meson is not its own antiparticle (the K^0 carries a non-zero value for the approximately conserved quantity known as strangeness) the particles with definite decay rates would be not K^0 or $\bar{K}^0$, but the linear combinations $K^0 \pm \bar{K}^0$. This was originally explained in terms of C conservation, but with C not conserved in the weak interactions, the argument may be equally well be based on CP conservation.

If we arbitrarily define the phases in the CP operator and in the K^0 and $\bar{K}^0$ states so that

$$CP|K^0\rangle = |\bar{K}^0\rangle$$

and

$$CP|\bar{K}^0\rangle = |K^0\rangle$$

then we can define self-charge-conjugate one-particle states

$$|K_1^0\rangle \equiv \frac{1}{\sqrt{2}}\left[|K^0\rangle + |\bar{K}^0\rangle\right]$$

and

$$|K_2^0\rangle \equiv \frac{1}{\sqrt{2}}\left[|K^0\rangle - |\bar{K}^0\rangle\right],$$

which have CP eigenvalues $+1$ and -1, respectively. The fastest available decay mode of these particles is into two-pion states, but CP conservation would allow this only[5] for the K_1, not the K_2. The K_2^0 would thus be expected to decay only by slower modes, into three pions or into a pion, muon or electron, and neutrino. Nevertheless, it was found by Fitch and Cronin in 1964 that the long-lived neutral K-meson does have a small probability for decaying into two pions [17]. The conclusion was that CP is not exactly conserved in the weak interactions, although it seems more nearly conserved than C or P individually.

As we shall see in Chapter 5, there are good reasons to believe that although neither C nor CP is strictly conserved, the product CPT is exactly conserved in all interactions, at least in any quantum field theory. It is CPT that provides a precise correspondence between particles and antiparticles, and in particular it is the fact that CPT commutes with the Hamiltonian that tells us that stable particles and antiparticles have exactly the same mass. Because CPT is antiunitary, it relates the S-matrix for an arbitrary process to the S-matrix for the *inverse* process with all spin three-components reversed and particles replaced with antiparticles. However, in cases where the S-matrix can be divided into a weak term $S^{(1)}$ that produces a given reaction and a strong term $S^{(0)}$ that acts in the initial and final states, we can use the same arguments that were used above in studying the implications of T conservation to show that the rate for any process is equal to the rate for the same process with particles replaced with antiparticles and spin three-components reversed, provided that we sum over sets of initial and final states that are complete with respect to $S^{(0)}$. In particular, although the partial rates for decay of the particle into a pair of final states β_1, β_2 with $S^{(0)}_{\beta_1\beta_2} \neq 0$ may differ from the partial rates for the decay of the antiparticle into the corresponding final states $CPT\beta_1$ and $CPT\beta_2$, we shall see in Section 3.5 that (without any approximations) the total decay rate of any particle is equal to that of its antiparticle.

We can now understand why the 1957 experiments on parity violation could be interpreted in the context of the existing theory of weak interactions as evidence that C conservation as well as P conservation are badly violated but CP is not. These theories

[5] The neutral K-mesons have spin zero, so the two-pion final state has $\ell = 0$, and hence $P = +1$. Further, $C = +1$ for two π^0s because the π^0 has $C = +1$, and also for an $\ell = 0$ $\pi^+\pi^-$ state because C interchanges the two pions.

were field theories, and therefore automatically conserved CPT. Since the experiments showed that PT conservation but not T conservation is badly violated in nuclear beta decay, *any* theory that was consistent with these experiments and in which CPT is conserved would have to also incorporate C but not CP non-conservation. Similarly, the observation in 1964 of small violations of CP conservation in the weak interactions together with the assumed invariance of all interactions under CPT allowed the immediate inference that the weak interactions also do not exactly conserve T. This has since been verified [18] by more detailed studies of the K^0–$\bar{K}^0$ system, but it has so far been impossible to find other direct evidence of the failure of invariance under time-reversal.

3.4 Rates and Cross-Sections

The S-matrix $S_{\beta\alpha}$ is the probability amplitude for the transition $\alpha \to \beta$, but what does this have to do with the transition rates and cross-sections measured by experimentalists? In particular, (3.3.2) shows that $S_{\beta\alpha}$ has a factor $\delta^4(p_\beta - p_\alpha)$, which ensures the conservation of the total energy and momentum, so what are we to make of the factor $[\delta^4(p_\beta - p_\alpha)]^2$ in the transition probability $|S_{\beta\alpha}|^2$? The proper way to approach these problems is by studying the way that experiments are actually done, using wave packets to represent particles localized far from each other before a collision, and then following the time-history of these superpositions of multi-particle states. In what follows we will instead give a quick and easy derivation of the main results, actually more a mnemonic than a derivation, with the excuse that (as far as I know) no interesting open problems in physics hinge on getting the fine points right regarding these matters.

We consider our whole system of physical particles to be enclosed in a large box with a macroscopic volume V. For instance, we can take this box as a cube, but with points on opposite sides identified, so that the single-valuedness of the spatial wave function requires the momenta to be quantized

$$\mathbf{p} = \frac{2\pi}{L}(n_1, n_2, n_3), \tag{3.4.1}$$

where the n_i are integers, and $L^3 = V$. Then all three-dimensional delta-functions become

$$\delta^3_V(\mathbf{p}' - \mathbf{p}) \equiv \frac{1}{(2\pi)^3}\int_V d^3x\, e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}} = \frac{V}{(2\pi)^3}\delta_{\mathbf{p}',\mathbf{p}}, \tag{3.4.2}$$

where $\delta_{\mathbf{p}',\mathbf{p}}$ is an ordinary Kronecker delta symbol, equal to one if the subscripts are equal and zero otherwise. The normalization condition (3.1.2) thus implies that the states we have been using have scalar products in a box which are not just sums of products of Kronecker deltas, but also contain a factor $[V/(2\pi)^3]^N$, where N is the number of particles in the state. To calculate transition probabilities we should use states of unit norm, so let us introduce states normalized approximately for our box

$$|\alpha\rangle^{\text{Box}} \equiv \left[(2\pi)^3/V\right]^{N_\alpha/2}|\alpha\rangle, \tag{3.4.3}$$

with norm

$${}^{\text{Box}}\langle\beta \mid \alpha\rangle^{\text{Box}} = \delta_{\beta\alpha}, \tag{3.4.4}$$

where $\delta_{\beta\alpha}$ is a product of Kronecker deltas, one for each three-momentum, spin, and species label, plus terms with particles permuted. Correspondingly, the S-matrix may be written

$$S_{\beta\alpha} = \left[V/(2\pi)^3\right]^{(N_\beta+N_\alpha)/2} S^{\mathrm{Box}}_{\beta\alpha}, \tag{3.4.5}$$

where $S^{\mathrm{Box}}_{\beta\alpha}$ is calculated using the states (3.4.3).

Of course, if we just leave our particles in the box forever, then every possible transition will occur again and again. To calculate a meaningful transition probability we also have to put our system in a 'time box'. We suppose that the interaction is turned on for only a time T. One immediate consequence is that the energy-conservation delta function is replaced with

$$\delta_T(E_\alpha - E_\beta) = \frac{1}{2\pi}\int_{-T/2}^{T/2} \exp\big(i(E_\alpha - E_\beta)t\big)\,dt. \tag{3.4.6}$$

The probability that a multi-particle system, which is in a state α before the interaction is turned on, is found in a state β after the interaction is turned off, is

$$P(\alpha\to\beta) = |S^{\mathrm{Box}}_{\beta\alpha}|^2 = \left[(2\pi)^3/V\right]^{N_\beta+N_\alpha}|S_{\beta\alpha}|^2. \tag{3.4.7}$$

This is the probability for a transition into one specific box state β. The number of one-particle box states in a momentum-space volume d^3p is $V d^3p/(2\pi)^3$, because this is the number of triplets of integers n_1, n_2, n_3 for which the momentum (3.4.1) lies in the momentum-space volume d^3p around $\mathbf{p}$. We shall define the final-state interval $d\beta$ as a product of d^3p for each final particle, so the total number of states in this range is

$$d\mathcal{N}_\beta = \left[V/(2\pi)^3\right]^{N_\beta} d\beta. \tag{3.4.8}$$

Hence the total probability for the system to wind up in a range $d\beta$ of final states is

$$dP(\alpha\to\beta) = P(\alpha\to\beta)d\mathcal{N}_\beta = \left[(2\pi)^3/V\right]^{N_\alpha}|S_{\beta\alpha}|^2 d\beta. \tag{3.4.9}$$

We will restrict our attention throughout this section to final states β that are not only different (however slightly) from the initial state α, but that also satisfy the more stringent condition, that no subset of the particles in the state β (other than the whole state itself) has precisely the same four-momentum as some corresponding subset of the particles in the state α. (In the language to be introduced in the next chapter, this means that we are considering only the connected part of the S-matrix.) For such states, we may define a delta function-free matrix element $M_{\beta\alpha}$:

$$S_{\beta\alpha} = -2\pi i\,\delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)\delta_T(E_\beta - E_\alpha)M_{\beta\alpha}. \tag{3.4.10}$$

Our introduction of the box allows us to interpret the squares of delta functions in $|S_{\beta\alpha}|^2$ for $\beta\neq\alpha$ as

$$\begin{aligned}\left[\delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)\right]^2 &= \delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)\delta^3_V(0) = \delta^3_V(\mathbf{p}_\beta - \mathbf{p}_\alpha)V/(2\pi)^3,\\ \left[\delta_T(E_\beta - E_\alpha)\right]^2 &= \delta_T(E_\beta - E_\alpha)\delta_T(0) = \delta_T(E_\beta - E_\alpha)T/2\pi,\end{aligned}$$

so Eq. (3.4.9) gives a differential transition probability

$$
\begin{aligned}
dP(\alpha \to \beta) = (2\pi)^2 \left[(2\pi)^3/V\right]^{N_\alpha - 1} (T/2\pi)|M_{\beta\alpha}|^2 \\
\times\, \delta_V^3(\mathbf{p}_\beta - \mathbf{p}_\alpha)\delta_T(E_\beta - E_\alpha)d\beta.
\end{aligned}
$$

If we let V and T be very large, the delta function product here may be interpreted as an ordinary four-dimensional delta function $\delta^4(p_\beta - p_\alpha)$. In this limit, the transition probability is simply proportional to the time T during which the interaction is acting, with a coefficient that may be interpreted as a differential transition *rate*:

$$
\begin{aligned}
d\Gamma(\alpha \to \beta) &\equiv dP(\alpha \to \beta)/T \\
&= (2\pi)^{3N_\alpha - 2} V^{1-N_\alpha} |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta,
\end{aligned}
\tag{3.4.11}
$$

where now

$$
S_{\beta\alpha} = -2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}. \tag{3.4.12}
$$

This is the master formula which is used to interpret calculations of S-matrix elements in terms of predictions for actual experiments. We will come back to the interpretation of the factor $\delta^4(p_\alpha - p_\beta)d\beta$ later in this section.

There are two cases of special importance:

$\boldsymbol{N_\alpha = 1}$ **:**

Here the volume V cancels in Eq. (3.4.11), which gives the transition rate for a single-particle state α to decay into a general multi-particle state β as

$$
d\Gamma(\alpha \to \beta) = 2\pi |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta. \tag{3.4.13}
$$

Of course, this makes sense only if the time T during which the interaction acts is much less than the mean lifetime τ_α of the particle α, so we cannot pass to the limit $T \to \infty$ in $\delta_T(E_\alpha - E_\beta)$. There is an unremovable width $\Delta E \simeq 1/T \gtrsim 1/\tau_\alpha$ in this delta function, so Eq. (3.4.13) is only useful if the total decay rate $1/\tau_\alpha$ is much less than any of the characteristic energies of the process.

$\boldsymbol{N_\alpha = 2}$ **:**

Here the rate (3.4.11) is proportional to $1/V$, or in other words, to the density of either particle at the position of the other one. Experimentalists generally report not the transition rate per density, but the *rate per flux*, also known as the *cross-section.* The flux of either particle at the position of the other particle is defined as the product of the density $1/V$ and the relative velocity u_α:

$$
\Phi_\alpha = u_\alpha / V. \tag{3.4.14}
$$

(A general definition of u_α is given below; for the moment we will content ourselves with specifying that if either particle is at rest then u_α is defined as the velocity of the other.) Thus the differential cross-section is

$$
d\sigma(\alpha \to \beta) \equiv d\Gamma(\alpha \to \beta)/\Phi_\alpha = (2\pi)^4 u_\alpha^{-1} |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha) d\beta. \tag{3.4.15}
$$

Even though the cases $N_\alpha = 1$ and $N_\alpha = 2$ are the most important, transition rates for $N_\alpha \geq 3$ are all measurable in principle, and some of them are very important in chemistry, astrophysics, etc. (For instance, in one of the main reactions that release energy in the sun, two protons and an electron turn into a deuteron and a neutrino.) Section 3.6 presents an application of the master transition rate formula (3.4.11) for general numbers N_α of initial particles.

We next take up the question of the Lorentz transformation properties of rates and cross-sections, which will help us to give a more general definition of the relative velocity u_α in Eq. (3.4.15). The Lorentz transformation rule (3.3.1) for the S-matrix is complicated by the momentum-dependent matrices associated with each particle's spin. To avoid this complication, consider the absolute-value squared of (3.3.1) (after factoring out the Lorentz-invariant delta function in Eq. (3.4.12)), and sum over all spins. The unitarity of the matrices $D^{(j)}(W)$ (or their analogs for zero mass) then shows that, apart from the energy factors in (3.3.1), the sum is Lorentz-invariant. That is, the quantity

$$\sum_{\text{spins}} |M_{\beta\alpha}|^2 \prod_\beta E \prod_\alpha E \equiv R_{\beta\alpha} \tag{3.4.16}$$

is a scalar function of the four-momenta of the particles in states α and β. (By $\prod_\alpha E$ and $\prod_\beta E$ is meant the product of all the single-particle energies $p^0 = \sqrt{\mathbf{p}^2 + m^2}$ for the particles in the states α and β.)

We can now write the spin-summed single-particle decay rate (3.4.13) as

$$\sum_{\text{spins}} d\Gamma(\alpha \to \beta) = 2\pi E_\alpha^{-1} R_{\beta\alpha} \delta^4(p_\beta - p_\alpha) d\beta / \prod_\beta E.$$

The factor $d\beta / \prod_\beta E$ may be recognized as the product of the Lorentz-invariant momentum-space volume elements (2.5.15), so it is Lorentz-invariant. So also are $R_{\beta\alpha}$ and $\delta^4(p_\beta - p_\alpha)$, leaving just the non-invariant factor $1/E_\alpha$, where E_α is the energy of the single initial particle. Our conclusion then is that the decay rate has the same Lorentz transformation property as $1/E_\alpha$. This is, of course, just the usual special-relativistic time dilation—the faster the particle, the slower it decays.

Similarly, our result (3.4.15) for the spin-summed cross-section may be written as

$$\sum_{\text{spins}} d\sigma(\alpha \to \beta) = (2\pi)^4 u_\alpha^{-1} E_1^{-1} E_2^{-1} R_{\beta\alpha} \delta^4(p_\alpha - p_\beta) d\beta / \prod_\beta E,$$

where E_1 and E_2 are the energies of the two particles in the initial state α. It is conventional to define the cross-section to be (when summed over spins) a Lorentz-invariant function of four-momenta. The factors $R_{\beta\alpha}$, $\delta^4(p_\beta - p_\alpha)$, and $d\beta / \prod_\beta E$ are already Lorentz-invariant, so this means that we must define the relative velocity u_α in arbitrary inertial frames so that $u_\alpha E_1 E_2$ is a scalar. We also mentioned earlier that in the Lorentz frame in which one particle (say, particle 1) is at rest, u_α is the velocity of the other particle. This uniquely determines u_α to have the value in general Lorentz frames

$$u_\alpha = \sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} / E_1 E_2, \tag{3.4.17}$$

where p_1, p_2 and m_1, m_2 are the four-momenta and mass of the two particles in the initial state α.

As a bonus, we note that in the 'center-of-mass' frame, where the total three-momentum vanishes, we have

$$p_1 = (E_1, \mathbf{p}), \qquad p_2 = (E_2, -\mathbf{p}),$$

and here Eq. (3.4.17) gives

$$u_\alpha = \frac{|\mathbf{p}|(E_1 + E_2)}{E_1 E_2} = \left| \frac{\mathbf{p}_1}{E_1} - \frac{\mathbf{p}_2}{E_2} \right| \tag{3.4.18}$$

as might have been expected for a relative velocity. However, in this frame u_α is not really a physical velocity; in particular, Eq. (3.4.18) shows that for extremely relativistic particles, it can take values as large as 2. [6]

We now turn to the interpretation of the so-called phase-space factor $\delta^4(p_\beta - p_\alpha)d\beta$, which appears in the general formula (3.4.11) for transition rates, and also in Eqs. (3.4.13) and (3.4.15) for decay rates and cross-sections. We here specialize to the case of the 'center-of-mass' Lorentz frame, where the total three-momentum of the initial state vanishes

$$\mathbf{p}_\alpha = 0. \tag{3.4.19}$$

(For $N_\alpha = 1$, this is just the case of a particle decaying at rest.) If the final state consists of particles with momenta $\mathbf{p}'_1, \mathbf{p}'_2, \ldots$, then

$$\delta^4(p_\beta - p_\alpha)d\beta = \delta^3(\mathbf{p}'_1 + \mathbf{p}'_2 + \cdots)\delta(E'_1 + E'_2 + \cdots - E)d^3p'_1 d^3p'_2 \cdots , \tag{3.4.20}$$

where $E \equiv E_\alpha$ is the total energy of the initial state. Any one of the $\mathbf{p}'_k$ integrals, say over $\mathbf{p}'_1$, can be done trivially by just dropping the momentum delta function

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta(E'_1 + E'_2 + \cdots - E)d^3p'_2 \cdots \tag{3.4.21}$$

with the understanding that wherever $\mathbf{p}'_1$ appears (as in E'_1) it must be replaced with

$$\mathbf{p}'_1 = -\mathbf{p}'_2 - \mathbf{p}'_3 - \cdots . \tag{3.4.22}$$

We can similarly use the remaining delta function to eliminate any one of the remaining integrals.

In the simplest case, there are just two particles in the final state. Here (3.4.21) gives

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta(E'_1 + E'_2 - E)d^3p'_2.$$

In more detail, this is

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \delta\left(\sqrt{|\mathbf{p}'_1|^2 + m'^2_1} + \sqrt{|\mathbf{p}'_1|^2 + m'^2_2} - E\right)|\mathbf{p}'_1|^2 d|\mathbf{p}'_1| d\Omega, \tag{3.4.23}$$

where $\mathbf{p}'_2 = -\mathbf{p}'_1$ and $d\Omega \equiv \sin\theta \, d\theta \, d\phi$ is the solid-angle differential for $\mathbf{p}'_1$. This can be simplified by using the standard formula

$$\delta(f(x)) = \delta(x - x_0)/|f'(x_0)|,$$

[6] Eq. (3.4.17) makes it obvious that $E_1 E_2 u_\alpha$ is a scalar. Also, when particle 1 is at rest, we have $\mathbf{p}_1 = 0$, $E_1 = m_1$, so $p_1 \cdot p_2 = -m_1 E_2$, and so Eq. (3.4.17) gives $u_\alpha = \sqrt{E_2^2 - m_2^2}/E_2 = |\mathbf{p}_2|/E_2$, which is just the velocity of particle 2.

where $f(x)$ is an arbitrary real function with a single simple zero at $x = x_0$. In our case, the argument $E'_1 + E'_2 - E$ of the delta function in Eq. (3.4.23) has a unique zero at $|\mathbf{p}'_1| = k'$, where

$$k' = \frac{\sqrt{(E^2 - m_1'^2 - m_2'^2)^2 - 4m_1'^2 m_2'^2}}{2E}, \tag{3.4.24}$$

$$E'_1 = \sqrt{k'^2 + m_1'^2} = \frac{E^2 - m_2'^2 + m_1'^2}{2E}, \tag{3.4.25}$$

$$E'_2 = \sqrt{k'^2 + m_2'^2} = \frac{E^2 - m_1'^2 + m_2'^2}{2E}, \tag{3.4.26}$$

with derivative

$$\left[\frac{d}{d|\mathbf{p}'_1|}\left(\sqrt{|\mathbf{p}'_1|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_1|^2 + m_2'^2} - E\right)\right]_{|\mathbf{p}'_1| = k'} = \frac{k'}{E'_1} + \frac{k'}{E'_2} = \frac{k'E}{E'_1 E'_2}. \tag{3.4.27}$$

We can thus drop the delta function and the differential $d|\mathbf{p}'_1|$ in Eq. (3.4.23), by dividing by (3.4.27),

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow \frac{k'E'_1E'_2}{E}\, d\Omega \tag{3.4.28}$$

with the understanding that k', E'_1, and E'_2 are given everywhere by Eqs. (3.4.24)–(3.4.26). In particular, the differential rate (3.4.13) for decay of a one-particle state of zero momentum and energy E into two particles is

$$\frac{d\Gamma(\alpha \to \beta)}{d\Omega} = \frac{2\pi k' E'_1 E'_2}{E} |M_{\beta\alpha}|^2 \tag{3.4.29}$$

and the differential cross-section for the two-body scattering process $12 \to 1'2'$ is given by Eq. (3.4.15) as

$$\frac{d\sigma(\alpha \to \beta)}{d\Omega} = \frac{(2\pi)^4 k' E'_1 E'_2}{E u_\alpha} |M_{\beta\alpha}|^2 = \frac{(2\pi)^4 k' E'_1 E'_2 E_1 E_2}{E^2 k} |M_{\beta\alpha}|^2, \tag{3.4.30}$$

where $k \equiv |\mathbf{p}_1| = |\mathbf{p}_2|$.

The above case $N_\beta = 2$ is particularly simple, but there is one nice result for $N_\beta = 3$ that is also worth recording. For $N_\beta = 3$, Eq. (3.4.21) gives

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow d^3p'_2 d^3p'_3 \times \delta\left(\sqrt{|\mathbf{p}'_2 + \mathbf{p}'_3|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_2|^2 + m_2'^2} + \sqrt{|\mathbf{p}'_3|^2 + m_3'^2} - E\right).$$

We write the momentum-space volume as

$$d^3p'_2 d^3p'_3 = |\mathbf{p}'_2|^2 d|\mathbf{p}'_2| |\mathbf{p}'_3|^2 d|\mathbf{p}'_3| d\Omega_3 d\phi_{23} d\cos\theta_{23},$$

where $d\Omega_3$ is the differential element of solid angle for $\mathbf{p}'_3$, and θ_{23} and ϕ_{23} are the polar and azimuthal angles of $\mathbf{p}'_2$ relative to the $\mathbf{p}'_3$ direction. The orientation of the plane spanned by $\mathbf{p}'_2$ and $\mathbf{p}'_3$ is specified by ϕ_{23} and the direction of $\mathbf{p}'_3$, with the remaining angle θ_{23} fixed by the energy-conservation condition

$$\sqrt{|\mathbf{p}'_2|^2 + 2|\mathbf{p}'_2||\mathbf{p}'_3|\cos\theta_{23} + |\mathbf{p}'_3|^2 + m_1'^2} + \sqrt{|\mathbf{p}'_2|^2 + m_2'^2} + \sqrt{|\mathbf{p}'_3|^2 + m_3'^2} = E.$$

The derivative of the argument of the delta function with respect to $\cos\theta_{23}$ is

$$\frac{\partial E_1'}{\partial\cos\theta_{23}} = \frac{|\mathbf{p}_2'||\mathbf{p}_3'|}{E_1'},$$

so we can do the integral over $\cos\theta_{23}$ by just dropping the delta function and dividing by this derivative

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow |\mathbf{p}_2'|d|\mathbf{p}_2'||\mathbf{p}_3'|d|\mathbf{p}_3'|E_1'd\Omega_3 d\phi_{23}.$$

Replacing momenta with energies, this is finally

$$\delta^4(p_\beta - p_\alpha)d\beta \longrightarrow E_1'E_2'E_3'\, dE_2'dE_3'd\Omega_3 d\phi_{23}. \tag{3.4.31}$$

But recall that the quantity (3.4.16), obtained by summing $|M_{\beta\alpha}|^2$ over spins and multiplying with the product of energies, is a scalar function of four-momenta. If we approximate this scalar as a constant, then Eq. (3.4.31) tells us that for a fixed initial state, the distribution of events plotted in the E_2', E_3' plane is uniform. Any departure from a uniform distribution of events in this plot thus provides a useful clue to the dynamics of the decay process, including possible centrifugal barriers or resonant intermediate states. This is known as a Dalitz plot [19], because of its use by Dalitz in 1953 to analyze the decay $K^+ \to \pi^+ + \pi^+ + \pi^-$.

3.5 Perturbation Theory

The technique that has historically been most useful in calculating the S-matrix is perturbation theory, an expansion in powers of the interaction term H_{int} in the Hamiltonian $H = H_0 + H_{\text{int}}$. Eqs. (3.2.7) and (3.1.18) give the S-matrix as

$$S_{\beta\alpha} = \delta(\beta-\alpha) - 2\pi i\delta(E_\beta - E_\alpha)T^+_{\beta\alpha}, \qquad T^+_{\beta\alpha} = \langle\beta\|H_{\text{int}}\|\alpha\rangle_{\text{in}},$$

where $|\alpha\rangle_{\text{in}}$ satisfies the Lippmann–Schwinger equation (3.1.17):

$$|\alpha\rangle_{\text{in}} = |\alpha\rangle + \int d\gamma\, \frac{T^+_{\gamma\alpha}|\gamma\rangle}{E_\alpha - E_\gamma + i\epsilon}.$$

Operating on this equation with H_{int} and taking the scalar product with $|\beta\rangle$ yields an integral equation for T^+

$$T^+_{\beta\alpha} = (H_{\text{int}})_{\beta\alpha} + \int d\gamma\, \frac{(H_{\text{int}})_{\beta\gamma}T^+_{\gamma\alpha}}{E_\alpha - E_\gamma + i\epsilon}, \tag{3.5.1}$$

where

$$(H_{\text{int}})_{\beta\alpha} \equiv \langle\beta\|H_{\text{int}}\|\alpha\rangle. \tag{3.5.2}$$

The perturbation series for $T^+_{\beta\alpha}$ is obtained by iteration from Eq. (3.5.1)

$$\begin{aligned} T^+_{\beta\alpha} = (H_{\text{int}})_{\beta\alpha} &+ \int d\gamma\, \frac{(H_{\text{int}})_{\beta\gamma}(H_{\text{int}})_{\gamma\alpha}}{E_\alpha - E_\gamma + i\epsilon} \\ &+ \int d\gamma\, d\gamma'\, \frac{(H_{\text{int}})_{\beta\gamma}(H_{\text{int}})_{\gamma\gamma'}(H_{\text{int}})_{\gamma'\alpha}}{(E_\alpha - E_\gamma + i\epsilon)(E_\alpha - E_{\gamma'} + i\epsilon)} + \cdots . \end{aligned} \tag{3.5.3}$$

The method of calculation based on Eq. (3.5.3), which dominated calculations of the S-matrix in the 1930s, is today known as *old-fashioned perturbation theory.* Its obvious drawback is that the energy denominators obscure the underlying Lorentz invariance of the S-matrix. It still has some uses, however, in clarifying the way that singularities of the S-matrix arise from various intermediate states. For the most part in this book, we will rely on a rewritten version of Eq. (3.5.3), known as *time-dependent perturbation theory*, which has the virtue of making Lorentz invariance much more transparent, while somewhat obscuring the contribution of individual intermediate states.

The easiest way to derive the time-ordered perturbation expansion is to use Eq. (3.2.5), which gives the S-operator as

$$S = U(\infty, -\infty),$$

where

$$U(\tau, \tau_0) \equiv \exp(iH_0\tau)\exp\big(-iH(\tau-\tau_0)\big)\exp(-iH_0\tau_0).$$

Differentiating this formula for $U(\tau, \tau_0)$ with respect to τ gives the differential equation

$$i\frac{d}{d\tau}U(\tau, \tau_0) = H_I(\tau)U(\tau, \tau_0), \tag{3.5.4}$$

where

$$H_I(t) \equiv \exp(iH_0 t)H_{\rm int}\exp(-iH_0 t). \tag{3.5.5}$$

(Operators with this sort of time-dependence are said to be defined in the *interaction picture*, to distinguish their time-dependence from the time-dependence

$$O_H(t) = \exp(iHt)O_H\exp(-iHt)$$

required in the Heisenberg picture of quantum mechanics.) Eq. (3.5.4) as well as the initial condition $U(\tau_0, \tau_0) = \mathbf{1}$ is obviously satisfied by the solution of the integral equation

$$U(\tau, \tau_0) = \mathbf{1} - i\int_{\tau_0}^{\tau} dt\, H_I(t)U(t, \tau_0). \tag{3.5.6}$$

By iteration of this integral equation, we obtain an expansion for $U(\tau, \tau_0)$ in powers of $H_{\rm int}$

$$\begin{aligned} U(\tau, \tau_0) = \mathbf{1} &- i\int_{\tau_0}^{\tau} dt_1\, H_I(t_1) + (-i)^2\int_{\tau_0}^{\tau} dt_1\int_{\tau_0}^{t_1} dt_2\, H_I(t_1)H_I(t_2) \\ &+ (-i)^3\int_{\tau_0}^{\tau} dt_1\int_{\tau_0}^{t_1} dt_2\int_{\tau_0}^{t_2} dt_3\, H_I(t_1)H_I(t_2)H_I(t_3) + \cdots . \end{aligned} \tag{3.5.7}$$

Setting $\tau = \infty$ and $\tau_0 = -\infty$ then gives the perturbation expansion for the S-operator:

$$\begin{aligned} S = \mathbf{1} &- i\int_{-\infty}^{\infty} dt_1\, H_I(t_1) + (-i)^2\int_{-\infty}^{\infty} dt_1\int_{-\infty}^{t_1} dt_2\, H_I(t_1)H_I(t_2) \\ &+ (-i)^3\int_{-\infty}^{\infty} dt_1\int_{-\infty}^{t_1} dt_2\int_{-\infty}^{t_2} dt_3\, H_I(t_1)H_I(t_2)H_I(t_3) + \cdots . \end{aligned} \tag{3.5.8}$$

This can also be derived directly from the old-fashioned perturbation expansion (3.5.3), by using the Fourier representation of the energy factors in Eq. (3.5.3):

$$(E_\alpha - E_\gamma + i\epsilon)^{-1} = -i\int_0^\infty d\tau\, \exp\bigl(i(E_\alpha - E_\gamma)\tau\bigr) \tag{3.5.9}$$

with the understanding that such integrals are to be evaluated by inserting a convergence factor $e^{-\epsilon\tau}$ in the integrand, with $\epsilon \to 0+$.

There is a way of rewriting Eq. (3.5.8) that proves very useful in carrying out manifestly Lorentz-invariant calculations. Define the *time-ordered product* of any time-dependent operators as the product with factors arranged so that the one with the latest time-argument is placed leftmost, the next-latest next to the leftmost, and so on. For instance,

$$T\{H_I(t)\} = H_I(t),$$

$$T\{H_I(t_1)H_I(t_2)\} = \theta(t_1 - t_2)H_I(t_1)H_I(t_2) + \theta(t_2 - t_1)H_I(t_2)H_I(t_1),$$

and so on, where $\theta(\tau)$ is the step function, equal to $+1$ for $\tau > 0$ and to zero for $\tau < 0$. The time-ordered product of n H_Is is a sum over all $n!$ permutations of the H_Is, each of which gives the same integral over all $t_1 \cdots t_n$, so Eq. (3.5.8) may be written

$$S = \mathbf{1} + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int_{-\infty}^{\infty} dt_1 dt_2 \cdots dt_n\, T\{H_I(t_1)\cdots H_I(t_n)\}. \tag{3.5.10}$$

This is sometimes known as the Dyson series [20]. This series can be summed if the $H_I(t)$ at different times all commute; the sum is then

$$S = \exp\left(-i\int_{-\infty}^{\infty} dt\, H_I(t)\right).$$

Of course, this is not usually the case; in general (3.5.10) does not even converge, and is at best an asymptotic expansion in whatever coupling constant factors appear in H_{int}. However Eq. (3.5.10) is sometimes written in the general case as

$$S = T\exp\left(-i\int_{-\infty}^{\infty} dt\, H_I(t)\right)$$

with T indicating here that the expression is to be evaluated by time-ordering each term in the series expansion for the exponential.

We can now readily find one large class of theories for which the S-matrix is manifestly Lorentz-invariant. Since the elements of the S-matrix are the matrix elements of the S-operator between free-particle states $|\alpha\rangle$, $|\beta\rangle$, etc., what we want is that the S-operator should commute with the operator $U_0(\Lambda, a)$ that produces Lorentz transformations on these free-particle states. Equivalently, the S-operator must commute with the generators of $U_0(\Lambda, a)$: H_0, $\mathbf{P}_0$, $\mathbf{J}_0$, and $\mathbf{K}_0$. To satisfy this requirement, let's try the hypothesis that $H_I(t)$ is an integral over three-space

$$H_I(t) = \int d^3x\, \mathcal{H}_I(\mathbf{x}, t) \tag{3.5.11}$$

with $\mathcal{H}_I(x)$ a scalar in the sense that

$$U_0(\Lambda, a)\mathcal{H}_I(x)U_0^{-1}(\Lambda, a) = \mathcal{H}_I(\Lambda x + a). \tag{3.5.12}$$

(By equating the coefficients of a^0 for infinitesimal transformations it can be checked that $\mathcal{H}_I(x)$ has a time-dependence consistent with Eq. (3.5.5).) Then S may be written as a sum of four-dimensional integrals

$$S = \mathbf{1} + \sum_{n=1}^{\infty} \frac{(-i)^n}{n!} \int d^4x_1 \cdots d^4x_n \, T\{\mathcal{H}_I(x_1) \cdots \mathcal{H}_I(x_n)\}. \tag{3.5.13}$$

Everything is now manifestly Lorentz invariant, except for the time-ordering of the operator product.

Now, the time-ordering of two spacetime points x_1, x_2 is Lorentz-invariant unless $x_1 - x_2$ is space-like, i.e., unless $(x_1 - x_2)^2 > 0$, so the time-ordering in Eq. (3.5.13) introduces no special Lorentz frame if (though not only if) the $\mathcal{H}_I(x)$ all commute at space-like or light-like separations:

$$[\mathcal{H}_I(x), \mathcal{H}_I(x')] = 0 \qquad \text{for} \qquad (x - x')^2 \geq 0. \tag{3.5.14}$$

We can use the results of Section 3.3 to give a formal non-perturbative proof that an interaction (3.5.11) satisfying Eqs. (3.5.12) and (3.5.14) does lead to an S-matrix with the correct Lorentz transformation properties. For an infinitesimal boost, Eq. (3.5.12) gives

$$i[\mathbf{K}_0, \mathcal{H}_I(\mathbf{x}, t)] = t\boldsymbol{\nabla}\mathcal{H}_I(\mathbf{x}, t) + \mathbf{x}\frac{\partial}{\partial t}\mathcal{H}_I(\mathbf{x}, t), \tag{3.5.15}$$

so integrating over $\mathbf{x}$ and setting $t = 0$,

$$[\mathbf{K}_0, H_{\text{int}}] = \left[\mathbf{K}_0, \int d^3x \, \mathcal{H}_I(\mathbf{x}, 0)\right] = [H_0, \mathbf{W}], \tag{3.5.16}$$

where

$$\mathbf{W} \equiv \int d^3x \, \mathbf{x}\mathcal{H}_I(\mathbf{x}, 0). \tag{3.5.17}$$

If (as is usually the case) the matrix elements of $\mathcal{H}_I(\mathbf{x}, 0)$ between eigenstates of H_0 are smooth functions of the energy eigenvalues, then the same is true of H_{int}, as is necessary for the validity of scattering theory, and also true of $\mathbf{W}$, which is necessary in the proof of Lorentz invariance. The other condition for Lorentz invariance, the commutation relation (3.3.21), is also valid if and only if

$$0 = [\mathbf{W}, H_{\text{int}}] = \int d^3x \int d^3y \, \mathbf{x}[\mathcal{H}_I(\mathbf{x}, 0), \mathcal{H}_I(\mathbf{y}, 0)]. \tag{3.5.18}$$

This condition would follow from the 'causality' condition (3.5.14), but provides a somewhat less restrictive sufficient condition for Lorentz invariance of the S-matrix.

Theories of this class are not the only ones that are Lorentz invariant, but the most general Lorentz invariant theories are not very different. In particular, there is always a commutation condition something like (3.5.14) that needs to be satisfied. This condition has no counterpart for non-relativistic systems, for which time-ordering is always Galilean-invariant. *It is this condition that makes the combination of Lorentz invariance and quantum mechanics so restrictive.* [7]

* * *

The methods described so far in this section are useful when the interaction operator $H_{\rm int}$ is sufficiently small. There is also a modified version of this approximation, known as the *distorted-wave Born approximation*, that is useful when the interaction contains two terms

$$H_{\rm int} = H_s + H_w \tag{3.5.19}$$

with H_w weak but H_s strong. We can define $|\alpha; s\rangle_{\rm in/out}$ as what the 'in' and 'out' states would be if H_s were the whole interaction

$$|\alpha; s\rangle_{\rm in/out} = |\alpha\rangle + (E_\alpha - H_0 \pm i\epsilon)^{-1} H_s |\alpha; s\rangle_{\rm in/out}. \tag{3.5.20}$$

We can then write (3.1.16) as

$$\begin{aligned}
T^+_{\beta\alpha} &= \langle\beta\|H_{\rm int}\|\alpha\rangle_{\rm in} \\
&= \left({}_{\rm out}\langle\beta; s| - {}_{\rm out}\langle\beta; s|H_s(E_\beta - H_0 + i\epsilon)^{-1}\right)(H_s + H_w)|\alpha\rangle_{\rm in} \\
&= {}_{\rm out}\langle\beta; s\|H_w\|\alpha\rangle_{\rm in} \\
&\quad + {}_{\rm out}\langle\beta; s\|[H_s - H_s(E_\beta - H_0 + i\epsilon)^{-1}(H_s + H_w)]\|\alpha\rangle_{\rm in},
\end{aligned}$$

and so

$$T^+_{\beta\alpha} = {}_{\rm out}\langle\beta; s\|H_w\|\alpha\rangle_{\rm in} + {}_{\rm out}\langle\beta; s\|H_s\|\alpha\rangle. \tag{3.5.21}$$

The second term on the right-hand side is just what $T^+_{s,\beta\alpha}$ would be in the presence of strong interactions alone

$$T^+_{s,\beta\alpha} = \langle\beta\|H_s\|\alpha; s\rangle_{\rm in} = {}_{\rm out}\langle\beta; s\|H_s\|\alpha\rangle. \tag{3.5.22}$$

(For a proof of Eq. (3.5.22), just drop H_w everywhere in the derivation of Eq. (3.5.21).) Eq. (3.5.21) is most useful when this second term vanishes: that is, when the process $\alpha \to \beta$ cannot be produced by the strong interactions alone. (For example, in nuclear beta decay we need a weak nuclear force to turn neutrons into protons, even though we cannot ignore the presence of the strong nuclear force acting in the initial and final nuclear states.) For such processes, the matrix element (3.5.22) vanishes, so Eq. (3.5.21) reads

$$T^+_{\beta\alpha} = {}_{\rm out}\langle\beta; s\|H_w\|\alpha\rangle_{\rm in}. \tag{3.5.23}$$

So far, this is all exact. However, this way of rewriting the T-matrix becomes worthwhile when H_w is so weak that we may neglect its effect on the state $|\alpha\rangle_{\rm in}$ in Eq. (3.5.23), and hence replace $|\alpha\rangle_{\rm in}$ with the state $|\alpha; s\rangle_{\rm in}$, which takes account only of the strong interaction H_s. In this approximation, Eq. (3.5.23) becomes

$$T^+_{\beta\alpha} \simeq {}_{\rm out}\langle\beta; s\|H_w\|\alpha; s\rangle_{\rm in}. \tag{3.5.24}$$

[7] We write the condition on x and x' here as $(x - x')^2 \geq 0$ instead of $(x - x')^2 > 0$, because as we shall see in Chapter 6, Lorentz invariance can be disturbed by troublesome singularities at $x = x'$.

This is valid to first order in H_w, but to all orders in H_s. This approximation is ubiquitous in physics; for instance, the S-matrix element for nuclear beta or gamma decay is calculated using Eq. (3.5.24) with H_s the strong nuclear interaction and H_w respectively either the weak nuclear interaction or the electromagnetic interaction, and with $|\beta; s\rangle_{\rm out}$ and $|\alpha; s\rangle_{\rm in}$ the final and initial nuclear states.

3.6 Implications of Unitarity

The unitarity of the S-matrix imposes an interesting and useful condition relating the amplitude $M_{\alpha\alpha}$ for forward scattering in an arbitrary multi-particle state α to the total rate for all reactions in that state. Recall that in the general case, where the state β may or may not be the same as the state α, the S-matrix may be written as in (3.3.2):

$$S_{\beta\alpha} = \delta(\beta - \alpha) - 2\pi i \delta^4(p_\beta - p_\alpha) M_{\beta\alpha}.$$

The unitarity condition then gives

$$\begin{aligned}
\delta(\gamma - \alpha) &= \int d\beta\, S^*_{\beta\gamma} S_{\beta\alpha} \\
&= \delta(\gamma - \alpha) - 2\pi i \delta^4(p_\gamma - p_\alpha) M_{\gamma\alpha} \\
&\quad + 2\pi i \delta^4(p_\gamma - p_\alpha) M^*_{\alpha\gamma} + 4\pi^2 \int d\beta\, \delta^4(p_\beta - p_\gamma)\delta^4(p_\beta - p_\alpha) M^*_{\beta\gamma} M_{\beta\alpha}.
\end{aligned}$$

Cancelling the term $\delta(\gamma - \alpha)$ and a factor $2\pi\delta^4(p_\gamma - p_\alpha)$, we find that for $p_\gamma = p_\alpha$

$$0 = -iM_{\gamma\alpha} + iM^*_{\alpha\gamma} + 2\pi \int d\beta\, \delta^4(p_\beta - p_\alpha) M^*_{\beta\gamma} M_{\beta\alpha}. \tag{3.6.1}$$

This is most useful in the special case $\alpha = \gamma$, where it reads

$$\mathrm{Im}\, M_{\alpha\alpha} = -\pi \int d\beta\, \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2. \tag{3.6.2}$$

Using Eq. (3.4.11), this can be expressed as a formula for the total rate for all reactions produced by an initial state α in a volume V

$$\begin{aligned}
\Gamma_\alpha &\equiv \int d\beta\, \frac{d\Gamma(\alpha \to \beta)}{d\beta} \\
&= (2\pi)^{3N_\alpha - 2} V^{1-N_\alpha} \int d\beta\, \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2 \\
&= -\frac{1}{\pi} (2\pi)^{3N_\alpha - 2} V^{1-N_\alpha}\, \mathrm{Im}\, M_{\alpha\alpha}.
\end{aligned} \tag{3.6.3}$$

In particular, where α is a two-particle state, this can be written

$$\mathrm{Im}\, M_{\alpha\alpha} = -u_\alpha \sigma_\alpha / 16\pi^3, \tag{3.6.4}$$

where u_α is the relative velocity (3.4.17) in state α, and σ_α is the *total* cross-section in this state, given by (3.4.15) as

$$\sigma_\alpha \equiv \int d\beta\, d\sigma(\alpha \to \beta)/d\beta = (2\pi)^4 u_\alpha^{-1} \int d\beta\, |M_{\beta\alpha}|^2 \delta^4(p_\beta - p_\alpha). \tag{3.6.5}$$

This is usually expressed in a slightly different way, in terms of a *scattering amplitude* $f(\alpha \to \beta)$. Eq. (3.4.30) shows that the differential cross-section for *two-body* scattering in the center-of-mass frame is

$$\frac{d\sigma(\alpha \to \beta)}{d\Omega} = \frac{(2\pi)^4 k' E_1' E_2' E_1 E_2}{kE^2} |M_{\beta\alpha}|^2, \tag{3.6.6}$$

where k' and k are the magnitudes of the momenta in the initial and final states. We therefore define the scattering amplitude as[8]

$$f(\alpha \to \beta) \equiv -\frac{4\pi^2}{E}\sqrt{\frac{k' E_1' E_2' E_1 E_2}{k}}\, M_{\beta\alpha}, \tag{3.6.7}$$

so that the differential cross-section is simply

$$\frac{d\sigma(\alpha \to \beta)}{d\Omega} = |f(\alpha \to \beta)|^2. \tag{3.6.8}$$

In particular, for *elastic* two-body scattering, we have

$$f(\alpha \to \beta) \equiv -\frac{4\pi^2 E_1 E_2}{E} M_{\beta\alpha}. \tag{3.6.9}$$

Using (3.4.18) for the relative velocity u_α, the unitarity prediction (3.6.3) now reads

$$\operatorname{Im} f(\alpha \to \alpha) = \frac{k}{4\pi}\sigma_\alpha. \tag{3.6.10}$$

This form of the unitarity condition (3.6.3) is known as the *optical theorem* [22].

There is a pretty consequence of the optical theorem that tells much about the pattern of scattering at high energy. The scattering amplitude f may be expected to be a smooth function of angle, so there must be some solid angle $\Delta\Omega$ within which $|f|^2$ has nearly the same value (say, within a factor 2) as in the forward direction. The total cross-section is then bounded by

$$\sigma_\alpha \geq \int |f|^2 d\Omega \geq \frac{1}{2}|f(\alpha \to \alpha)|^2 \Delta\Omega \geq \frac{1}{2}|\operatorname{Im} f(\alpha \to \alpha)|^2 \Delta\Omega.$$

Using Eq. (3.6.10) then yields an upper bound on $\Delta\Omega$

$$\Delta\Omega \leq \frac{32\pi^2}{k^2 \sigma_\alpha}. \tag{3.6.11}$$

As we shall see in the next section, total cross-sections are usually expected to approach constants or grow slowly at high energy, so Eq. (3.6.11) shows that the solid angle around the forward direction within which the differential cross-section is roughly constant shrinks at least as fast as $1/k^2$ for $k \to \infty$. This increasingly narrow peak in the forward direction at high energies is known as a *diffraction peak*.

[8] The phase of f is conventional, and is motivated by the wave mechanical interpretation [21] of f as the coefficient of the outgoing wave in the solution of the time-independent Schrödinger equation. The normalization of f used here is slightly unconventional for inelastic scattering; usually f is defined so that a ratio of final and initial velocities appears in the formula for the differential cross-section.

Returning now to the general case of reactions involving arbitrary numbers of particles, we can use Eq. (3.6.2) together with CPT invariance to say something about the relations for total interaction rates of particles and antiparticles. Because CPT is antiunitary, its conservation does not in general imply any simple relation between a process $\alpha \to \beta$ and the process with particles replaced with their antiparticles. Instead, it provides a relation between a process and the *inverse* process involving antiparticles: we can use the same arguments that allowed us to deduce (3.3.46) from time-reversal invariance to show that CPT invariance requires the S-matrix to satisfy the condition

$$S_{\beta,\alpha} = S_{\mathcal{CPT}\alpha,\mathcal{CPT}\beta}, \tag{3.6.12}$$

where $\mathcal{CPT}$ indicates that we must reverse all spin z-components, change all particles into their corresponding antiparticles, and multiply the matrix element by various phase factors for the particles in the initial state and by their complex conjugates for the particles in the final state. Since CPT invariance also requires particles to have the same masses as their corresponding antiparticles, the same relation holds for the coefficient of $\delta^4(p_\alpha - p_\beta)$ in $S_{\beta\alpha}$:

$$M_{\beta,\alpha} = M_{\mathcal{CPT}\alpha,\mathcal{CPT}\beta}. \tag{3.6.13}$$

In particular, when the initial and final states are the same the phase factors all cancel, and Eq. (3.6.13) says that

$$\begin{aligned} &M_{p_1\sigma_1 n_1;\, p_2\sigma_2 n_2;\, \ldots,\, p_1\sigma_1 n_1;\, p_2\sigma_2 n_2;\, \ldots} \\ &\qquad = M_{p_1,-\sigma_1,n_1^c;\, p_2,-\sigma_2,n_2^c;\, \ldots,\, p_1,-\sigma_1,n_1^c;\, p_2,-\sigma_2,n_2^c;\, \ldots}, \end{aligned} \tag{3.6.14}$$

where a superscript c on n indicates the antiparticle of n. The generalized optical theorem (3.6.2) then tells us that *the total reaction rate from an initial state consisting of some set of particles is the same as for an initial state consisting of the corresponding antiparticles with spins reversed*:

$$\Gamma_{p_1\sigma_1 n_1;\, p_2\sigma_2 n_2;\, \ldots} = \Gamma_{p_1,-\sigma_1,n_1^c;\, p_2,-\sigma_2,n_2^c;\, \ldots}. \tag{3.6.15}$$

In particular, applying this to one-particle states, we see that the decay rate of any particle equals the decay rate of the antiparticle with reversed spin. Rotational invariance does not allow particle decay rates to depend on the spin z-component of the decaying particle, so a special case of the general result (3.6.15) is that unstable particles and their corresponding antiparticles have precisely the same lifetimes.

* * *

The same argument that led from the unitarity condition $S^\dagger S = \mathbf{1}$ to our result (3.6.2) also allows us to use the other unitarity relation $SS^\dagger = \mathbf{1}$ to derive the result

$$\operatorname{Im} M_{\alpha\alpha} = -\pi \int d\beta\, \delta^4(p_\beta - p_\alpha) |M_{\alpha\beta}|^2. \tag{3.6.16}$$

Putting this together with Eq. (3.6.2) then yields the reciprocity relation

$$\int d\beta\, \delta^4(p_\beta - p_\alpha) |M_{\beta\alpha}|^2 = \int d\beta\, \delta^4(p_\beta - p_\alpha) |M_{\alpha\beta}|^2 \tag{3.6.17}$$

or in other words

$$\int d\beta\, \frac{d\Gamma(\alpha\to\beta)}{d\beta} = \int d\beta\, \frac{d\Gamma(\beta\to\alpha)}{d\alpha}. \tag{3.6.18}$$

This result can be used in deriving some of the most important results of kinetic theory [23]. If $P_\alpha d\alpha$ is the probability of finding the system in a volume $d\alpha$ of the space of multi-particle states $|\alpha\rangle$, then the rate of decrease in P_α due to transitions to all other states is $P_\alpha \int d\beta\, d\Gamma(\alpha\to\beta)/d\beta$, while the rate of increase of P_α due to transitions from all other states is $\int d\beta\, P_\beta\, d\Gamma(\beta\to\alpha)/d\alpha$; the rate of change of P_α is then

$$\frac{dP_\alpha}{dt} = \int d\beta\, P_\beta \frac{d\Gamma(\beta\to\alpha)}{d\alpha} - P_\alpha \int d\beta\, \frac{d\Gamma(\alpha\to\beta)}{d\beta}. \tag{3.6.19}$$

It follows immediately that $\int P_\alpha d\alpha$ is time-independent. (Just interchange the labelling of the integration variables in the integral in the second term in Eq. (3.6.19).) On the other hand, the rate of change of the entropy $-\int d\alpha\, P_\alpha \ln P_\alpha$ is

$$-\frac{d}{dt}\int d\alpha\, P_\alpha \ln P_\alpha = -\int d\alpha \int d\beta\, (\ln P_\alpha + 1)\left[P_\beta \frac{d\Gamma(\beta\to\alpha)}{d\alpha} - P_\alpha \frac{d\Gamma(\alpha\to\beta)}{d\beta}\right].$$

Interchanging the labelling of the integration variables in the second term, this may be written

$$-\frac{d}{dt}\int d\alpha\, P_\alpha \ln P_\alpha = \int d\alpha \int d\beta\, P_\beta \ln\left(\frac{P_\beta}{P_\alpha}\right) \frac{d\Gamma(\beta\to\alpha)}{d\alpha}.$$

Now, for any positive probabilities P_α and P_β, the function $P_\beta \ln(P_\beta/P_\alpha)$ satisfies the inequality[9]

$$P_\beta \ln\left(\frac{P_\beta}{P_\alpha}\right) \geq P_\beta - P_\alpha.$$

The rate of change of the entropy is thus bounded by

$$-\frac{d}{dt}\int d\alpha\, P_\alpha \ln P_\alpha \geq \int d\alpha \int d\beta\, [P_\beta - P_\alpha] \frac{d\Gamma(\beta\to\alpha)}{d\alpha}$$

or interchanging variables of integration in the second term

$$-\frac{d}{dt}\int d\alpha\, P_\alpha \ln P_\alpha \geq \int d\alpha \int d\beta\, P_\beta \left[\frac{d\Gamma(\beta\to\alpha)}{d\alpha} - \frac{d\Gamma(\alpha\to\beta)}{d\beta}\right].$$

But the unitarity relation (3.6.18) tells us that the integral over α on the right-hand side vanishes, so we may conclude that the entropy always increases:

$$-\frac{d}{dt}\int d\alpha\, P_\alpha \ln P_\alpha \geq 0. \tag{3.6.20}$$

This is the 'Boltzmann H-theorem.' This theorem is often derived in statistical mechanics textbooks either by using the Born approximation, for which $|M_{\beta\alpha}|^2$ is symmetric in α and β so that $d\Gamma(\beta\to\alpha)/d\alpha = d\Gamma(\alpha\to\beta)/d\beta$, or by assuming time-reversal invariance,

[9] The difference between the left- and right-hand sides approaches the positive quantity $(P_\alpha - P_\beta)^2/2P_\beta$ for $P_\alpha \to P_\beta$, and has a derivative with respect to P_α that is positive- or negative-definite for all $P_\alpha > P_\beta$ or $P_\alpha < P_\beta$, respectively.

which would tell us that $|M_{\beta\alpha}|^2$ is unchanged if we interchange α and β and also reverse all momenta and spins. Of course, neither the Born approximation nor time-reversal invariance are exact, so it is a good thing that the unitarity result (3.6.18) is all we need in order to derive the H-theorem.

The increase of the entropy stops when the probability P_α becomes a function only of conserved quantities such as the total energy and charge. In this case the conservation laws require $d\Gamma(\beta \to \alpha)/d\alpha$ to vanish unless $P_\alpha = P_\beta$, so we can replace P_β with P_α in the first term of Eq. (3.6.19). Using Eq. (3.6.18) again then shows that in this case, P_α is time-independent. Here again, we need only the unitarity relation (3.6.18), not the Born approximation or time-reversal invariance.

3.7 Partial-Wave Expansions[10]

It is often convenient to work with the S-matrix in a basis of free-particle states in which all variables are discrete, except for the total momentum and energy. This is possible because the components of the momenta $\mathbf{p}_1, \ldots, \mathbf{p}_n$ in an n-particle state of definite total momentum $\mathbf{p}$ and total energy E form a $(3n-4)$-dimensional compact space; for instance, for $n = 2$ particles in the center-of-mass frame with $\mathbf{p} = 0$, this space is a two-dimensional spherical surface. Any function on such a compact space may be expanded in a series of generalized 'partial waves', such as the spherical harmonics that are commonly used in representing functions on the two-sphere. We may thus define a basis for these n-particle states that apart from the continuous variables $\mathbf{p}$ and E is discrete: we label the free-particle states in such a basis as $|E, \mathbf{p}, N\rangle$, with the index N incorporating all spin and species labels as well as whatever indices are used to label the generalized partial waves. These states may conveniently be chosen to be normalized so that their scalar products are:

$$\langle E', \mathbf{p}', N' \mid E, \mathbf{p}, N\rangle = \delta(E' - E)\delta^3(\mathbf{p}' - \mathbf{p})\delta_{N'N}. \tag{3.7.1}$$

The S-operator then has matrix elements in this basis of the form

$$\langle E', \mathbf{p}', N'|S|E, \mathbf{p}, N\rangle = \delta(E' - E)\delta^3(\mathbf{p}' - \mathbf{p})S_{N'N}(E, \mathbf{p}), \tag{3.7.2}$$

where $S_{N'N}$ is a unitary matrix. Similarly, the T-operator, whose free-particle matrix elements $\langle\beta|T|\alpha\rangle$ are defined to be the quantities $T^+_{\beta\alpha}$ defined by Eq. (3.1.18), may be expressed in our new basis (in accordance with Eq. (3.4.12)) as

$$\langle E', \mathbf{p}', N'|T|E, \mathbf{p}, N\rangle = \delta^3(\mathbf{p}' - \mathbf{p})M_{N'N}(E, \mathbf{p}) \tag{3.7.3}$$

and the relation (3.2.7) is now an ordinary matrix equation:

$$S_{N'N}(E, \mathbf{p}) = \delta_{N'N} - 2\pi i M_{N'N}(E, \mathbf{p}). \tag{3.7.4}$$

We shall use this general formalism in the following section; for the present we will concentrate on reactions in which the initial state involves just two particles.

[10] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

For example, consider a state consisting of two non-identical particles of species n_1, n_2 with non-zero masses M_1, M_2 and arbitrary spins s_1, s_2. In this case, the states may be labelled by their total momentum $\mathbf{p} = \mathbf{p}_1 + \mathbf{p}_2$, the energy E, the species labels n_1, n_2, the spin z-components σ_1, σ_2, and a pair of integers ℓ, m (with $|m| \leq \ell$) that specify the dependence of the state on the directions of, say, $\mathbf{p}_1$. Alternatively, we can form a convenient discrete basis by using Clebsch–Gordan coefficients [9] to combine the two spins to give a total spin s with z-component μ, and then using Clebsch–Gordan coefficients again to combine this with the orbital angular momentum ℓ with three-component m to form a total angular momentum j with three-component σ. This gives a basis of states $|E, \mathbf{p}, j, \sigma, \ell, s, n\rangle$ (with n a 'channel index' labelling the two particle species n_1, n_2), defined by their scalar products with the states of definite individual momenta and spin three-components:

$$\langle \mathbf{p}_1, \sigma_1; \mathbf{p}_2, \sigma_2; n' \mid E, \mathbf{p}, j, \sigma, \ell, s, n\rangle = \left(\frac{|\mathbf{p}_1| E_1 E_2}{E}\right)^{-1/2} \delta^3(\mathbf{p} - \mathbf{p}_1 - \mathbf{p}_2)$$
$$\times \delta\left(E - \sqrt{\mathbf{p}_1^2 + M_1^2} - \sqrt{\mathbf{p}_2^2 + M_2^2}\right) \delta_{n'n} \sum_{m,\mu} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu) Y_\ell^m(\widehat{\mathbf{p}}_1). \tag{3.7.5}$$

where Y_ℓ^m are the usual spherical harmonics [24]. The factor $(|\mathbf{p}_1| E_1 E_2 / E)^{-1/2}$ is inserted so that in the center-of-mass frame these states will be properly normalized:

$$\langle E', \mathbf{p}', j', \sigma', \ell', s', n' \mid E, \mathbf{0}, j, \sigma, \ell, s, n\rangle$$
$$= \delta^3(\mathbf{p}') \delta(E' - E) \delta_{j'j} \delta_{\sigma'\sigma} \delta_{\ell'\ell} \delta_{s's} \delta_{n'n}. \tag{3.7.6}$$

For identical particles to avoid double counting we must integrate only over half of the two-particle momentum space, so an extra factor $\sqrt{2}$ should appear in the scalar product (3.7.6).

In the center-of-mass frame the matrix elements of any momentum-conserving and rotationally-invariant operator O must take the form:

$$\langle E, \mathbf{p}', j', \sigma', \ell', s', n' | O | E, \mathbf{0}, j, \sigma, \ell, s, n\rangle = \delta^3(\mathbf{p}') O^j_{\ell' s' n', \ell s n}(E) \delta_{j'j} \delta_{\sigma'\sigma}. \tag{3.7.7}$$

(The fact that this is diagonal in j and σ follows from the commutation of O with $\mathbf{J}^2$ and J_3, and the further fact that the coefficient of $\delta_{\sigma\sigma'}$ is independent of σ follows from the commutation of O with $J_1 \pm i J_2$. This is a special case of a general result known as the Wigner–Eckart theorem [25].) Applying this to the operator M whose matrix elements are the quantities $M_{\beta\alpha}$, it follows that the scattering amplitude (3.6.7) in the center-of-mass system takes the form

$$f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \to \mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n')$$
$$\equiv -4\pi^2 \sqrt{\frac{k' E_1' E_2' E_1 E_2}{E^2 k}} M_{\mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n}$$
$$= -\frac{4\pi^2}{k} \sum_{j\sigma\ell m\ell' m' s\mu s'\mu'} C_{s_1 s_2}(s, \mu; \sigma_1, \sigma_2) C_{\ell s}(j, \sigma; m, \mu)$$
$$\times C_{s_1' s_2'}(s', \mu'; \sigma_1', \sigma_2') C_{\ell' s'}(j, \sigma; m', \mu') Y_{\ell'}^{m'*}(\widehat{\mathbf{k}}') Y_\ell^m(\widehat{\mathbf{k}}) M^j_{\ell' s' n', \ell s n}(E). \tag{3.7.8}$$

The differential scattering cross-section is $|f|^2$. We will take the direction of the initial momentum $\mathbf{k}$ to be along the three-direction, in which case

$$Y_\ell^m(\widehat{\mathbf{k}}) = \delta_{m0}\sqrt{\frac{2\ell+1}{4\pi}}. \tag{3.7.9}$$

Integrating $|f|^2$ over the direction of the final momentum $\mathbf{k}'$ and respectively summing and averaging over final and initial spin three-components gives the total cross-section[11] for transitions from channel n to n':

$$\begin{aligned}\sigma(n\to n';E) = &\frac{\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s\ell' s'}(2j+1)\\ &\times\left|\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}(E)\right|^2.\end{aligned} \tag{3.7.10}$$

Summing (3.7.10) over all two-body channels gives the total cross-section for all elastic or inelastic two-body reactions:

$$\begin{aligned}\sum_{n'}\sigma(n\to n';E) = &\frac{\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)\\ &\times\left[(\mathbf{1}-S^j(E))^\dagger(\mathbf{1}-S^j(E))\right]_{\ell sn,\ell sn}.\end{aligned} \tag{3.7.11}$$

For comparison, Eqs. (3.7.8), (3.7.9), (3.7.4), and the Clebsch–Gordan sum rules give the spin-averaged forward scattering amplitude as

$$f(n;E) = \frac{i}{2k(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)[\mathbf{1}-S^j(E)]_{\ell sn,\ell sn}.$$

The optical theorem (3.6.10) then gives the total cross-section as

$$\sigma_{\text{total}}(n;E) = \frac{2\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)\,\mathrm{Re}[\mathbf{1}-S^j(E)]_{\ell sn,\ell sn}. \tag{3.7.12}$$

If only two-body channels can be reached from the channel n at energy E, then the matrix $S^j(E)$ (or at least some submatrix that includes channel n) is unitary, and thus

$$\left[(\mathbf{1}-S^j(E))^\dagger(\mathbf{1}-S^j(E))\right]_{\ell sn,\ell sn} = 2\,\mathrm{Re}[\mathbf{1}-S^j(E)]_{\ell sn,\ell sn}, \tag{3.7.13}$$

[11] In deriving this result, we use standard sum rules [9] for the Clebsch–Gordan coefficients: first

$$\sum_{\sigma_1\sigma_2} C_{s_1s_2}(s,\mu;\sigma_1,\sigma_2)C_{s_1s_2}(\bar s,\bar\mu;\sigma_1,\sigma_2) = \delta_{s\bar s}\delta_{\mu\bar\mu},$$

and the same with primes; then

$$\sum_{m\sigma} C_{\ell s}(j,\sigma;m,\bar\sigma)C_{\ell' s}(\bar j,\bar\sigma;m,\bar\sigma) = \delta_{j\bar j}\delta_{\sigma\bar\sigma},$$

and finally

$$\sum_{\sigma\mu} C_{\ell s}(j,\sigma;0,\mu)C_{\ell' s}(j,\sigma;0,\mu) = \frac{2j+1}{2\ell+1}\delta_{\ell\ell'}.$$

so (3.7.12) and (3.7.11) are equal. On the other hand, if channels involving three or more particles are open, then the difference of (3.7.12) and (3.7.11) gives the total cross-section for producing extra particles:

$$\sigma_{\text{production}}(n;E) = \frac{\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)[\mathbf{1} - S^j(E)^\dagger S^j(E)]_{\ell sn,\ell sn}, \tag{3.7.14}$$

and this must be positive.

The partial wave expansion is particularly useful when applied to processes where the relevant part of the S-matrix is diagonal. This is the case, for instance, if the initial channel n contains just two spinless particles, and no other channels are open at this energy, as in $\pi^+ - \pi^+$ or $\pi^+ - \pi^0$ scattering at energies below the threshold for producing extra pions (provided one ignores weak and electromagnetic interactions). For a pair of spinless particles we have $j = \ell$, and angular momentum conservation keeps the S-matrix diagonal. It is also possible for the S-matrix to be diagonal in certain processes involving particles with spin; for instance in pion–nucleon scattering we can have $j = \ell + \frac{1}{2}$ or $j = \ell - \frac{1}{2}$, but for a given j these two states have opposite parity, so they cannot be connected by non-zero S-matrix elements. In any case, if for some n and E the S-matrix elements $S^j_{N',\ell sn}(E,\mathbf{0})$ all vanish unless N' is the two-body state j,ℓ,s,n, then unitarity requires that

$$S^j_{\ell' s' n',\ell sn}(E) = \exp[2i\delta_{j\ell sn}(E)]\delta_{\ell'\ell}\delta_{s's}\delta_{n'n}, \tag{3.7.15}$$

where $\delta_{j\ell sn}(E)$ is a real phase, commonly known as the *phase shift*. This formula is also often used where the two-body part of the S-matrix is diagonal but channels containing three or more particles are also open; in such cases the phase shift must have a positive imaginary part, to keep (3.7.14) positive. For real phase shifts, the elastic and total cross-sections are then given by Eq. (3.7.10) or Eq. (3.7.12) as:

$$\sigma(n \to n;E) = \sigma_{\text{total}}(n;E) = \frac{4\pi}{k^2(2s_1+1)(2s_2+1)}\sum_{j\ell s}(2j+1)\sin^2\delta_{j\ell sn}(E). \tag{3.7.16}$$

This familiar result is usually derived in non-relativistic quantum mechanics by studying the coordinate space wave function for a particle in a potential. The derivation given here is offered both to show that the partial wave expansion applies for elastic scattering even at relativistic velocities, and also to emphasize that it depends on no particular dynamical assumptions, only on unitarity and invariance principles.

It is also often useful to introduce phase shifts in dealing with problems where several channels are open, forming a few irreducible representations of some internal symmetry group. The classic example of such an internal symmetry is isotopic spin symmetry, for which the channel index n includes a specification of the isospins T_1, T_2 of the two particles together with their three-components t_1, t_2; the states in channel n may be expressed as linear combinations of the tth components of irreducible representations T, with coefficients given by the familiar Clebsch–Gordan coefficients $C_{T_1T_2}(T,t;t_1,t_2)$. Suppose that for the channels and energy of interest the S-matrix is diagonal in ℓ and s as well as j, T, and t. Unitarity and isospin symmetry then allow us to write the S-matrix as

$$S^j_{\ell' s' T' t',\ell sTt} = \exp[2i\delta_{j\ell sT}(E)]\delta_{\ell'\ell}\delta_{s's}\delta_{T'T}\delta_{t't}, \tag{3.7.17}$$

with $\delta_{j\ell sT}(E)$ a real phase shift, t-independent according to the Wigner–Eckart theorem. The partial cross-sections can again be calculated from Eq. (3.7.10), and the total cross-section is given by Eq. (3.7.12) as

$$\sigma_{\text{total}}(t_1, t_2; E) = \frac{4\pi}{k^2(2s_1+1)(2s_2+1)} \times \sum_{j\ell sTt} (2j+1) C_{T_1 T_2}(T, t; t_1, t_2)^2 \sin^2 \delta_{j\ell sT}(E). \tag{3.7.18}$$

For instance, in pion–pion scattering we have phase shifts $\delta_{\ell\ell 0T}(E)$ with $T = 0$ or $T = 2$ for each even ℓ and $T = 1$ for each odd ℓ, while for pion–nucleon scattering we have phase shifts $\delta_{j, j\pm\frac{1}{2}, \frac{1}{2}, T}$ with $T = \frac{1}{2}$ or $T = \frac{3}{2}$.

We can gain some useful insight about the threshold behavior of the scattering amplitudes and phase shifts from considerations of analyticity that are nearly independent of any dynamical assumptions. Unless there are special circumstances that would produce singularities in momentum space, we would expect the matrix element $M_{\mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n'; \mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n}$ to be an analytic function[12] of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k = 0$ or $k' = 0$ or (for elastic scattering) $k = k' = 0$. Turning to the partial wave expansion (3.7.8) for M, we note that $k^\ell Y_\ell^m(\widehat{\mathbf{k}})$ is a simple polynomial function of the three-vector $\mathbf{k}$, so in order for the matrix element to be an analytic function of the three-momenta $\mathbf{k}$ and $\mathbf{k}'$ near $k = 0$ or $k' = 0$, the coefficients $M^j_{\ell' s' n', \ell s n}$ or equivalently $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell' s' n', \ell s n}$ must go as $k^{\ell+1/2} k'^{\ell'+1/2}$ when k and/or k' go to zero. Hence for small k and/or k', it is only the lowest partial waves in the initial and/or final state that contribute appreciably to the scattering amplitude. We have three possible cases:

Exothermic reactions

Here k' approaches a finite value as $k \to 0$, and in this limit $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell' s' n', \ell s n}$ goes as $k^{\ell+1/2}$. The cross-section (3.7.11) in this case goes as $k^{2\ell-1}$, where ℓ is here the *lowest* orbital angular momentum that can lead to the reaction. In the most usual case $\ell = 0$, so the reaction cross-section goes as $1/k$. (This is the case, for instance, in the absorption of slow neutrons by complex nuclei, or for the annihilation of electron–positron pairs into photons at low energy, aside from the higher-order effects of Coulomb forces.) The reaction rate is the cross-section times the flux, which goes like k, so the rate for an exothermic reaction behaves like a constant for $k \to 0$. However, it is the cross-section rather than the reaction rate that determines the probability of absorption when a beam crosses a given thickness of target material, and the factor $1/k$ makes this probability very high for slow neutrons in an absorbing material like boron.

Endothermic Reactions

[12] For instance, in the Born approximation (3.2.8), M is proportional to the Fourier transform of the coordinate space matrix elements of the interaction, and hence is analytic at zero momenta as long as these matrix elements fall off sufficiently rapidly at large separations. The chief exception is for scattering involving long-distance forces, such as the Coulomb force.

Here the reaction is forbidden until k reaches a finite threshold value, where $k' = 0$. Just above this threshold $\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - S^j_{\ell's'n',\ell sn}$ goes as $(k')^{\ell'+1/2}$. The cross-section (3.7.11) in this case goes as $(k')^{2\ell'+1}$, where ℓ' is here the *lowest* orbital angular momentum that can be produced at threshold. In the most usual case $\ell' = 0$, so the reaction cross-section rises above threshold like k', and hence like $\sqrt{E - E_{\text{threshold}}}$. (This is the case, for instance, in the associated production of strange particles, or the production of electron–positron pairs in the scattering of photons.)

Elastic Reactions

Here $k = k'$, so k and k' go to zero together. (This is the case where $n' = n$, or where n' consists of particles in the same isotopic spin multiplets as are those in n.) In elastic scattering the partial waves with $\ell = \ell' = 0$ are always present, so in the limit $k \to 0$ the scattering amplitude (3.7.8) becomes a constant:

$$\begin{aligned} &f(\mathbf{k}\sigma_1, -\mathbf{k}\sigma_2, n \to \mathbf{k}'\sigma_1', -\mathbf{k}'\sigma_2', n') \\ &\qquad \to \sum_{s\sigma} C_{s_1 s_2}(s, \sigma; \sigma_1, \sigma_2) C_{s_1' s_2'}(s, \sigma; \sigma_1', \sigma_2') a_s(n \to n'), \end{aligned} \tag{3.7.19}$$

where a is a constant, known as the *scattering length*, defined by the limit

$$S^s_{0sn',0sn} \longrightarrow \delta_{n'n} + 2ika_s(n \to n') \qquad (k = k' \to 0). \tag{3.7.20}$$

Summing $4\pi|f|^2$ over final spins and averaging over initial spins gives the total cross-section for the transition $n \to n'$ at $k = k' = 0$:

$$\sigma(n \to n'; k = 0) = \frac{4\pi}{(2s_1+1)(2s_2+1)} \sum_s (2s+1)|a_s(n \to n')|^2. \tag{3.7.21}$$

The classic instance of the use of this formula is in neutron–proton scattering, where there are two scattering lengths, with the spin singlet length a_0 considerably larger than the spin triplet length a_1.

The partial wave expansion can also be used to make a crude guess about the behavior of cross-sections at high energy. With decreasing wavelengths, we may expect scattering to be described more or less classically: a particle of momentum k and orbital angular momentum ℓ would have an impact parameter ℓ/k, and will therefore strike a disk of radius R if $\ell \leq kR$. This can be interpreted as a statement about S-matrix elements:

$$S^j_{\ell's'n',\ell sn} \longrightarrow \begin{cases} 0, & \ell \ll kR_n, \\ 1, & \ell \gg kR_n, \end{cases} \tag{3.7.22}$$

where R_n is some sort of interaction radius for channel n. For a given $\ell \gg s$, there are $2s+1$ values of j, all close enough to ℓ to approximate $2j+1 \simeq 2\ell+1$, so the sum over j and s in Eq. (3.7.12) merely gives a factor of order

$$\sum_{js}(2j+1) = (2\ell+1)\sum_s(2s+1) = (2\ell+1)(2s_1+1)(2s_2+1).$$

The total cross-section is then given for $k \gg 1/R_n$ by Eq. (3.7.12) as

$$\sigma_{\text{total}}(n;E) \longrightarrow \frac{2\pi}{k^2} \sum_{\ell \lesssim kR_n} (2\ell+1) \longrightarrow 2\pi R_n^2. \tag{3.7.23}$$

In exactly the same way, Eq. (3.7.10) gives the elastic scattering cross-section

$$\sigma(n \to n; E) \longrightarrow \pi R_n^2. \tag{3.7.24}$$

The difference between Eqs. (3.7.23) and (3.7.24) gives an inelastic cross-section πR_n^2, which is what we would expect for collisions with an opaque disk of radius R_n. (The somewhat surprising elastic scattering cross-section πR_n^2 may be attributed to diffraction by the disk.) On the other hand, if we assume along with Eq. (3.7.22) that $S^j_{\ell' s' n', \ell s n}$ is complex only for impact parameters ℓ/k within a small range of width $\Delta_n \ll R_n$ around $\ell/k = R_n$, then using the inequality $|\,\mathrm{Im}(1 - S^j_{\ell' s' n', \ell s n})| \leq 2$, the same analysis gives a bound on the real part of the forward scattering amplitude

$$|\,\mathrm{Re}\, f(n;E)| \leq 2kR_n\Delta_n \ll |\,\mathrm{Im}\, f(n;E)|. \tag{3.7.25}$$

The smallness of the real part of the forward scattering amplitude at high energy is confirmed by experiment.

So far we have not said anything about whether the interaction radius R_n itself may depend on energy. As a very crude guess, we may take R_n as the distance at which the factor $\exp(-\mu r)$ in the Yukawa potential (1.2.74) takes a value proportional to some unknown power of E, in which case R_n goes as $\log E$ for $E \to \infty$, and the cross-sections go as $(\log E)^2$. As it happens, it has been rigorously shown [26] on the basis of very general assumptions that the total cross-section can grow no faster than $(\log E)^2$ for $E \to \infty$, and in fact the observed proton–proton total cross-section rises something like $(\log E)^2$ at high energies, so this rough picture of high-energy scattering does seem to have some correspondence with reality.

3.8 Resonances[13]

It often happens that the particles participating in a multi-particle collision can form an intermediate state consisting of a *single* unstable particle R, that eventually decays into the particles observed as the final state. If the total decay rate of R is small the cross-section exhibits a rapid variation (usually a peak), known as a *resonance*, at the energy of the intermediate state R.

We shall see that the behavior of the cross-sections near a resonance is pretty much prescribed by the unitarity condition alone, which is a good thing since there are a number of very different mechanisms that can produce a nearly stable state:

(a) The simplest possibility is that the Hamiltonian can be decomposed into two terms, a 'strong' Hamiltonian $H_0 + H_s$, which has the particle R as an eigenstate, plus a

[13] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

weak perturbation H_w, which allows R to decay into various states, including the initial and final states α, β of our collision process. For instance there is a neutral particle, the Z^0, with $j = 1$ and mass 91 GeV, that would be stable in the absence of the electroweak interactions. These interactions allow the Z^0 to decay into electron–positron pairs, muon–antimuon pairs, etc., but with a total decay rate that is much less than the Z^0 mass. In 1989 the Z^0 particle was seen as a resonance[14] in electron–positron collisions at CERN and Stanford, in the reactions $e^+ + e^- \to Z^0 \to e^+ + e^-$, $e^+ + e^- \to Z^0 \to \mu^+ + \mu^-$, etc.

(b) In some cases a particle is long-lived because there is a potential barrier that nearly prevents its constituents from escaping. The classic example is nuclear alpha decay: it may be energetically possible for a nucleus to emit an alpha particle (a He^4 nucleus) but the strong electrostatic repulsion between the alpha particle and the nucleus creates a barrier region around the daughter nucleus, which the alpha particle is classically forbidden to enter. The decay then can proceed only by quantum mechanical barrier penetration, and is exponentially slow. Such an unstable state shows up as a resonance in the scattering of the alpha particle on the daughter nucleus. For instance, the lowest-energy state of the Be^8 nucleus is unstable against decay into two alpha particles, and is seen as a resonance in He^4–He^4 scattering. (In addition to Coulomb barriers, there are also centrifugal barriers that help lengthen the life of alpha-, beta-, and gamma-unstable nuclei of high spin.)

(c) It is possible for complicated systems to be nearly unstable for statistical reasons, without the presence of any potential barriers or weak interactions. For instance, an excited state of a heavy nucleus may be able to decay only if, through a statistical fluctuation, a large part of its energy is concentrated on a single neutron. This state will then show up as a resonance in the scattering of a neutron on the daughter nucleus.

These mechanisms for producing long-lived states are so different that it is truly fortunate that most of the properties of resonances follow from unitarity alone, without regard to the dynamical mechanism that produces the resonance.

First, let's consider the energy dependence of the matrix element for a reaction near a resonance. A wave packet $\int d\alpha\, g(\alpha)|\alpha\rangle_{\rm in} \exp(-iE_\alpha t)$ of 'in' states has a time-dependence given by (3.1.19)

$$\int d\alpha\, g(\alpha)|\alpha\rangle_{\rm in} e^{-iE_\alpha t} = \int d\alpha\, g(\alpha)|\alpha\rangle_0 e^{-iE_\alpha t} + \int d\beta\, |\beta\rangle_0 \int d\alpha\, \frac{e^{-iE_\alpha t} g(\alpha) T^+_{\beta\alpha}}{E_\alpha - E_\beta + i\epsilon}.$$

As mentioned in Section 3.1, a pole in the function $T^+_{\beta\alpha}$ in the lower-half complex E_α plane would make a contribution to the second term that decays exponentially as $t \to \infty$. Specifically, a pole at $E_\alpha = E_R - i\Gamma/2$ yields a term in the *amplitude* that behaves like

[14] Incidentally, this example shows that a resonant state only needs to decay *relatively* slowly; the Z^0 lifetime is 2.6×10^{-25} seconds, which is not long enough for a Z^0 travelling near the speed of light to cross an atomic nucleus. What is important is that the decay rate is 36 times smaller than the rate $M_Z/\hbar$ of oscillation of the Z^0 wave function in its rest frame.

$\exp(-iE_R t - \Gamma t/2)$, so it corresponds to a state whose *probability* decays like $\exp(-\Gamma t)$. We conclude then that a long-lived state of energy E_R with a slow decay rate Γ produces a term in the scattering amplitude that varies as

$$T^+_{\beta\alpha} \sim (E_\alpha - E_R + i\Gamma/2)^{-1} + \text{constant}. \tag{3.8.1}$$

To go further, it will be convenient to adopt as a basis the orthonormal discrete multi-particle states $|\mathbf{p}, E, N\rangle$ discussed in the previous section; $\mathbf{p}$ and E are the total momentum and energy, and N is an index that takes only discrete (though infinitely many) values. In this basis, the S-matrix may be written

$$S_{\mathbf{p}'E'N',\mathbf{p}EN} = \delta^3(\mathbf{p}' - \mathbf{p})\delta(E' - E)S_{N'N}(\mathbf{p}, E). \tag{3.8.2}$$

Near a resonance, we expect the center-of-mass frame amplitude $S(\mathbf{0}, E) \equiv \mathcal{S}(E)$ to have the form

$$\mathcal{S}_{N'N}(E) \equiv S_{N'N}(\mathbf{0}, E) = \mathcal{S}_{0N'N} + \frac{\mathcal{R}_{N'N}}{E - E_R + i\Gamma/2}, \tag{3.8.3}$$

where $\mathcal{S}_0$ and $\mathcal{R}$ are approximately constant at least over the relatively small range of energies $|E - E_R| \lesssim \Gamma$.

In this basis, the unitarity of the S-matrix is an ordinary matrix equation

$$\mathcal{S}(E)^\dagger \mathcal{S}(E) = \mathbf{1}. \tag{3.8.4}$$

Applied to Eq. (3.8.3), this tells us that the non-resonant background S-matrix is unitary

$$\mathcal{S}_0^\dagger \mathcal{S}_0 = \mathbf{1}, \tag{3.8.5}$$

and also that the residue matrix $\mathcal{R}$ satisfies the two conditions

$$\mathcal{S}_0^\dagger \mathcal{R} - \mathcal{R}^\dagger \mathcal{S}_0 = 0, \tag{3.8.6}$$

$$-\frac{i}{2}\Gamma \mathcal{S}_0^\dagger \mathcal{R} + \frac{i}{2}\Gamma \mathcal{R}^\dagger \mathcal{S}_0 + \mathcal{R}^\dagger \mathcal{R} = 0. \tag{3.8.7}$$

These conditions can be put in a more transparent form by setting

$$\mathcal{R} \equiv -i\Gamma \mathcal{A} \mathcal{S}_0. \tag{3.8.8}$$

The unitarity conditions on the matrix $\mathcal{A}$ are then simply

$$\mathcal{A}^\dagger = \mathcal{A}, \qquad \mathcal{A}^2 = \mathcal{A}. \tag{3.8.9}$$

Any such Hermitian idempotent matrix is called a *projection matrix.* Such matrices can always be expressed as a sum of dyads of orthonormal vectors $u^{(r)}$:

$$\mathcal{A}_{N'N} = \sum_r u^{(r)}_{N'} u^{(r)*}_N, \qquad \sum_N u^{(r)*}_N u^{(s)}_N = \delta_{rs}. \tag{3.8.10}$$

The discrete part of the S-matrix is then

$$\mathcal{S}_{N'N}(E) = \sum_{N''} \left[\delta_{N'N''} - i\frac{\Gamma}{E - E_R + i\Gamma/2} \sum_r u^{(r)}_{N'} u^{(r)*}_{N''} \right] \mathcal{S}_{0N''N}. \tag{3.8.11}$$

Each term in the sum over r can be thought of as arising from a different resonant state, all these states having the same values for E_R and Γ.

What has this to do with rates and cross-sections? For simplicity let's now ignore the non-resonant background scattering, setting $\mathcal{S}_{0N'N}$ equal to $\delta_{N'N}$; we will come back to the more general case a little later. Then for the two-body discrete center-of-mass states described in the previous section, Eq. (3.8.11) reads:

$$\mathcal{S}_{j'\sigma'\ell's'n',j\sigma\ell sn}(E) = \delta_{j'j}\delta_{\sigma'\sigma}\delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - i\frac{\Gamma}{E-E_R+i\Gamma/2}\sum_r u^{(r)}_{j'\sigma'\ell's'n'}u^{(r)*}_{j\sigma\ell sn}. \tag{3.8.12}$$

In all cases the label r will include an index σ_R giving the z-component of the total angular momentum of the resonant state; for a resonant state of total angular momentum j_R, σ_R takes $2j_R+1$ values. If there is no other degeneracy, then r just labels the value of σ_R, and

$$u^{(\sigma_R)}_{j\sigma\ell sn} = \delta_{jj_R}\delta_{\sigma\sigma_R}u_{\ell sn}, \tag{3.8.13}$$

where $u_{\ell sn}$ are a set of complex amplitudes that (because of the Wigner–Eckart theorem) are independent of σ. Now Eq. (3.8.12) gives the amplitude S^j defined by Eq. (3.7.7) as

$$S^j_{\ell's'n',\ell sn}(E) = \delta_{\ell'\ell}\delta_{s's}\delta_{n'n} - i\delta_{jj_R}\frac{\Gamma}{E-E_R+i\Gamma/2}u_{\ell's'n'}u^*_{\ell sn}. \tag{3.8.14}$$

Also, Eq. (3.8.10) now reads

$$\sum_{\ell sn}|u_{\ell sn}|^2 + \cdots = 1 \tag{3.8.15}$$

with the dots representing the positive contribution of any states containing three or more particles. As we shall see, the quantities $|u_{\ell sn}|^2$ have the interpretation of branching ratios for the decay of the resonant state into the various accessible two-body states.

Eq. (3.7.12) now gives the total cross-section for all reactions in channel n:

$$\sigma_{\text{total}}(n;E) = \frac{\pi(2j_R+1)}{k^2(2s_1+1)(2s_2+1)}\frac{\Gamma\Gamma_n}{(E-E_R)^2+\Gamma^2/4}, \tag{3.8.16}$$

where

$$\Gamma_n \equiv \Gamma\sum_{\ell s}|u_{\ell sn}|^2. \tag{3.8.17}$$

This is a version of the celebrated Breit–Wigner single-level formula [27]. We can also use these results to calculate the cross-section for resonant scattering from an initial two-body channel n to a final two-body channel n'. Using Eq. (3.8.14) in Eq. (3.7.10) gives

$$\sigma(n\to n';E) = \frac{\pi(2j_R+1)}{k^2(2s_1+1)(2s_2+1)}\frac{\Gamma_n\Gamma_{n'}}{(E-E_R)^2+\Gamma^2/4}. \tag{3.8.18}$$

This shows that the probabilities that the resonant state will decay into any one of the final two-body channels n' are proportional to the $\Gamma_{n'}$. According to Eq. (3.8.15), the sum of the Γ_n (including contributions from final states containing three or more particles) is just equal to the total decay rate Γ, so we can conclude that Γ_n is just the rate for the decay of the resonant state into channel n.

We see in Eqs. (3.8.16) and (3.8.18) the characteristic resonant peak at energy E_R, with a width (the full width at half maximum) equal to the decay rate Γ. (The individual Γ_n are often called partial widths.) Since $\Gamma_n \leq \Gamma$, the total cross-section at the peak of the resonance is roughly bounded by one square wavelength, $(2\pi/k)^2$. This rule, that cross-sections at a single resonance are roughly bounded by a square wavelength, is universally applicable even in classical physics (where energy conservation plays the role played here by unitarity), as for instance in the resonant interaction of sound waves with bubbles in the sea, or gravitational waves with gravitational wave antennae. (In the latter case, the branching ratio for oscillations in any laboratory mass to lose their energy through gravitational radiation is tiny, so the cross-section even at a resonance peak is vastly less then a square wavelength [28].)

Incidentally, it often happens that a resonance is detected, but energy measurements are insufficiently precise to resolve its width. In this case, what is measured experimentally is the integral of the cross-section over the resonant peak. For the total cross-section (3.8.14), this is

$$\int \sigma_{\text{total}}(n; E)\, dE = \frac{2\pi^2(2j_R + 1)\Gamma_n}{k_R^2(2s_1 + 1)(2s_2 + 1)}. \tag{3.8.19}$$

Such experiments can reveal only the partial width for decay of the resonant state into the initial particles, not the total width or branching ratios.

This formalism can also be applied when the resonant states with a given spin z-component form a multiplet related by some symmetry group. For instance, to the extent that isospin symmetry is respected, for a resonance of total isospin T_R the index r labelling resonant states includes a specification not only of the angular-momentum z-component σ_R, but also of an isospin three-component t_R, taking values $-T_R, -T_R + 1, \ldots, T_R$. In this case there is no change in the above results for the total and partial cross-sections, because each two-body channel n has definite values t_1, t_2 of the isospin z-components of the two particles, and hence can only couple to the resonant state with the single value $t_1 + t_2$ for t_R. The partial widths Γ_n here depend on t_1 and t_2 only through factors $C_{T_1 T_2}(T_R, t_R; t_1, t_2)^2$.

The presence of a resonance shows itself in a characteristic behavior of phase shifts near the resonance. Returning to the general formula (3.8.11) (but still taking $\mathcal{S}_0 = \mathbf{1}$), we see from Eq. (3.8.10) that for each individual resonant state r, there is an eigenvector $u_N^{(r)}$ of $\mathcal{S}_{N'N}(E)$ with eigenvalue

$$\exp[2i\delta^{(r)}(E)] = 1 - i\frac{\Gamma}{E - E_R + i\Gamma/2}$$

or, in other words,

$$\tan \delta^{(r)}(E) = -\frac{\Gamma/2}{E - E_R}. \tag{3.8.20}$$

We see that over an energy range of order Γ centered around the resonant energy, the the 'eigenphase' $\delta^{(r)}(E)$ jumps from a value $\nu\pi$ (with ν a positive or negative integer) below the resonance to $(\nu + 1)\pi$ above it. However, in order to use this result to say something about reaction rates, we need to know the eigenvectors $u_N^{(r)}$, which, in general, have components with arbitrary numbers of particles with various momenta, spin, and species.

These results are much more useful in those special cases when the particles in a particular channel N are forbidden (usually by conservation laws) from making a transition to any other channel. With this assumption, it is not difficult to include the effects of a non-resonant background scattering matrix $\mathcal{S}_0$ in the general result (3.8.11). In order for $\mathcal{S}_{N'N}$ to vanish for some particular N and all $N' \neq N$, it is necessary that the same is true of $\mathcal{S}_{0N'N}$, and also true of $u_{N'}^{(r)}$ for any r for which $u_N^{(r)} \neq 0$. The unitarity requirement (3.8.5) then requires that for this N

$$\mathcal{S}_{0N'N} = \exp(2i\delta_{0N})\delta_{N'N}$$

and Eq. (3.8.10) requires that

$$u_N^{(r)*}u_N^{(s)} = \delta_{rs},$$

so that there can be only one term r in Eq. (3.8.11) for which $u_N^{(r)} \neq 0$. In this case, Eq. (3.8.11) gives

$$\mathcal{S}_{N'N}(E) = \delta_{N'N}\left[1 - \frac{i\Gamma}{E - E_R + i\Gamma/2}\right]\exp(2i\delta_{0N}) \equiv \delta_{N'N}\exp[2i\delta_N(E)]$$

with total phase shift

$$\delta_N(E) = \delta_{0N} - \arctan\left(\frac{\Gamma/2}{E - E_R}\right). \tag{3.8.21}$$

We see that over an energy range of order Γ centered around the resonant energy E_R, the phase shift $\delta_N(E)$ jumps from a value δ_{0N} below the

resonance to $\delta_{0N} + \pi$ above it. For instance, as we saw in the previous section, these assumptions are satisfied in various two-body reactions such as pion–pion and pion–nucleon scattering at energies below the threshold for producing extra pions, with N incorporating the total and orbital angular-momenta j, ℓ (with $j = \ell$ for pion–pion scattering) and the total angular-momentum z-component σ, as well as the total isospin T and its three-component t. The Wigner–Eckart theorem allows the phase shifts to depend only on j, ℓ, and T, not on t or σ. There are famous resonances in these channels: in pion–pion scattering there is a resonance at 770 MeV called ρ with $j = \ell = 1$, $T = 1$, and $\Gamma = 150$ MeV; in pion–nucleon scattering there is a resonance at 1232 MeV called Δ with $j = \frac{3}{2}$, $\ell = 1$, $T = \frac{3}{2}$ and $\Gamma = 110$ to 120 MeV.

Inspection of Eq. (3.7.12) or Eq. (3.7.18) shows that the total cross-section reaches a peak when the resonant phase shift passes through $\pi/2$ (or odd-integer multiples of $\pi/2$.) The non-resonant phase shifts are typically rather small, so as we saw earlier, σ_{total} will exhibit a sharp peak when the phase shift δ_ℓ goes through $\pi/2$, at an energy close to E_R. However, it sometimes happens that the non-resonant background phase shift δ_{0N} is near $\pi/2$, in which case the cross-section will exhibit a sharp *dip* as the phase shift rises through π near E_R, due to destructive interference between the resonance and the non-resonant background amplitude. Such dips were first observed by Ramsauer and Townsend [29] in 1922, in the scattering of electrons by noble gas atoms.

Problems

1. Consider a theory with a separable interaction; that is,

$$\langle\beta|H_{\rm int}|\alpha\rangle = g\, u_\beta u_\alpha^*,$$

where g is a real coupling constant, and u_α is a set of complex quantities with

$$\sum_\alpha |u_\alpha|^2 = 1.$$

Use the Lippmann–Schwinger equation (3.1.16) to find explicit solutions for the 'in' and 'out' states and the S-matrix.

2. Suppose that a resonance of spin one is discovered in e^+e^- scattering at a total energy of 150 GeV and with a cross-section (in the center-of-mass frame, averaged over initial spins, and summed over final spins) for elastic e^+e^- scattering at the peak of the resonance equal to $10^{-34}\,\text{cm}^2$. What is the branching ratio for the decay of the resonant state R by the mode $R \to e^- + e^+$? What is the total cross-section for e^+e^- scattering at the peak of the resonance? (In answering both questions, ignore the non-resonant background scattering.)

3. Express the differential cross-section for two-body scattering in the *laboratory* frame, in which one of the two particles is initially at rest, in terms of kinematic variables and the matrix element $M_{\beta\alpha}$. (Derive the result directly, without using the results derived in this chapter for the differential cross-section in the center-of-mass frame.)

4. Derive the perturbation expansion (3.5.8) directly from the expansion (3.5.3) of old-fashioned perturbation theory.

5. We can define 'standing wave' states $|\alpha\rangle^0$ by a modified version of the Lippmann–Schwinger equation

$$|\alpha\rangle^0 = |\alpha\rangle_0 + \text{P. V.}\, \frac{1}{E_\alpha - H_0} H_{\rm int}|\alpha\rangle^0.$$

Show that the matrix $K_{\beta\alpha} = \pi\delta(E_\beta - E_\alpha)\langle\beta|H_{\rm int}|\alpha\rangle^0$ is Hermitian. Show how to express the S-matrix in terms of the K-matrix.

6. Express the differential cross-section for elastic π^+–proton and π^-–proton scattering in terms of the phase shifts for states of definite total angular momentum, parity, and isospin.

7. Show that the states $|E, \mathbf{p}, j, \sigma, \ell, s, n\rangle$ defined by Eq. (3.7.5) are correctly normalized to have the scalar products (3.7.6).

References

1. For more details, see M. L. Goldberger and K. M. Watson, *Collision Theory* (John Wiley & Sons, New York, 1964); R. G. Newton, *Scattering Theory of Waves and Particles*, 2nd edn (Springer-Verlag, New York, 1982).

1a. B. Lippmann and J. Schwinger, *Phys. Rev.* **79**, 469 (1950).

2. J. A. Wheeler, *Phys. Rev.* **52**, 1107 (1937); W. Heisenberg, *Z. Phys.* **120**, 513, 673 (1943).

3. M. Born, *Z. Phys.* **37**, 863 (1926); **38**, 803 (1926).

4. C. Møller, *Kgl. Danske Videnskab. Mat. Fys. Medd.* **23**, No. 1 (1945); **22**, No. 19 (1946).

5. G. D. Rochester and C. C. Butler, *Nature* **160**, 855 (1947). For a historical review, see G. D. Rochester, in *Pions to Quarks – Particle Physics in the 1950s*, ed. by L. M. Brown, M. Dresden, and L. Hoddeson (Cambridge University Press, Cambridge, UK, 1989).

6. G. Breit, E. U. Condon, and R. S. Present, *Phys. Rev.* **50**, 825 (1936); B. Cassen and E. U. Condon, *Phys. Rev.* **50**, 846 (1936); G. Breit and E. Feenberg, *Phys. Rev.* **50**, 850 (1936).

7. M. A. Tuve, N. Heydenberg, and L. R. Hafstad, *Phys. Rev.* **50**, 806 (1936).

8. M. Gell-Mann, Cal. Tech. Synchotron Laboratory Report CTSL-20 (1961); *Phys. Rev.* **125**, 1067 (1962); Y. Ne'eman, *Nucl. Phys.* **26**, 222 (1961).

9. See e.g. A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, Princeton, 1957): Chapter 3 (where $C_{j_1 j_2}(jm; m_1 m_2)$ is denoted $(j_1 j_2 jm | j_1 m_1 j_2 m_2)$); M. E. Rose, *Elementary Theory of Angular Momentum* (John Wiley & Sons, New York, 1957): Chapter III (where $C_{j_1 j_2}(jm; m_1 m_2)$ is denoted $C(j_1 j_2 j; m_1 m_2 m)$).

10. G. Feinberg and S. Weinberg, *Nuovo Cimento* Serie X, **14**, 571 (1959).

11. W. Chinowsky and J. Steinberger, *Phys. Rev.* **95**, 1561 (1954); also see B. Ferretti, *Report of an International Conference on Fundamental Particles and Low Temperatures*, Cambridge, 1946 (The Physical Society, London, 1947).

12. T. D. Lee and C. N. Yang, *Phys. Rev.* **104**, 254 (1956).

13. C. S. Wu *et al.*, *Phys. Rev.* **105**, 1413 (1957).

14. R. Garwin, L. Lederman, and M. Weinrich, *Phys. Rev.* **105**, 1415 (1957); J. I. Friedman and V. L. Telegdi, *Phys. Rev.* **105**, 1681 (1957).

15. K. M. Watson, *Phys. Rev.* **88**, 1163 (1952).

16. M. Gell-Mann and A. Pais, *Phys. Rev.* **97**, 1387 (1955); also see A. Pais and O. Piccioni, *Phys. Rev.* **100**, 1487 (1955).

17. J. H. Christenson, J. W. Cronin, V. L. Fitch, and R. Turlay, *Phys. Rev. Letters* **13**, 138 (1964).

18. K. R. Schubert *et al.*, *Phys. Lett.* **31B**, 662 (1970). This reference analyzes neutral kaon data without the assumption of CPT invariance, and finds that the part of the CP-violating amplitude that conserves CPT and violates T has both real and imaginary parts five standard deviations from zero, while the part that conserves T and violates CPT is within a standard deviation of zero.

19. R. H. Dalitz, *Phil. Mag.* **44**, 1068 (1953); also see E. Fabri, *Nuovo Cimento* **11**, 479 (1954).

20. F. J. Dyson, *Phys. Rev.* **75**, 486, 1736 (1949).

21. See, e.g., L. I. Schiff, *Quantum Mechanics*, 1st edn (McGraw-Hill, New York, 1949): Section 19.

22. This was first proved in classical electrodynamics. See, e.g., H. A. Kramers, *Atti Congr. Intern. Fisici, Como*, 1927; reprinted in H. A. Kramers, *Collected Scientific Papers* (North-Holland, Amsterdam, 1956). For the proof in quantum mechanics, see E. Feenberg, *Phys. Rev.* **40**, 40 (1932); N. Bohr, R. E. Peierls, and G. Placzek, *Nature* **144**, 200 (1939).

23. A general version of this argument was given in the late 1960s in unpublished work of C. N. Yang and C. P. Yang. Also see A. Aharony, in *Modern Developments in Thermodynamics* (Wiley, New York, 1973): pp. 95–114, and references therein.

24. See, e.g., A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, Princeton, 1957): Chapter 2; M. E. Rose, *Elementary Theory of Angular Momentum* (John Wiley & Sons, New York, 1957): Appendix III; L. D. Landau and E. M. Lifshitz, *Quantum Mechanics – Non Relativistic Theory*, 3rd edn (Pergamon Press, Oxford, 1977): Section 28.

25. E. P. Wigner, *Gruppentheorie* (Friedrich Vieweg und Sohn, Braunschweig, 1931); C. Eckart, *Rev. Mod. Phys.* **2**, 305 (1930).

26. M. Froissart, *Phys. Rev.* **123**, 1053 (1961).

27. G. Breit and E. P. Wigner, *Phys. Rev.* **49**, 519 (1936).

28. See, e.g., S. Weinberg, *Gravitation and Cosmology* (Wiley, New York, 1972): Section 10.7.

29. R. Kollath, *Phys. Zeit.* **31**, 985 (1931).